The Canine Rosetta Stone

How Far Are We from Translating What Dogs Say?

A Survey of Cross-Species Communication Decoding in the Age of Foundation Models

The `woof-to-word` Project

`github.com/mcxiaoxiao/woof2word`

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Abstract

Consumer apps already claim to translate dog barks into sentences, and flagship projects apply foundation models to whales, dolphins, and songbirds. How far are we, really, from translating what a dog says? This survey answers through a five-rung *translation ladder*: signal detection, affect classification, intent and context inference, referential semantics, and open-ended dialogue. Synthesizing canine behavioral biology, the human machine-translation stack, cross-species decoding projects, and the commercial market, we locate today's best systems on the second rung, with scattered footholds on the third. Two findings shape the verdict. First, a *semantic gap*: dogs signal affect, arousal, identity, and some referential content, but show no evidence of the generative syntax that sentence-level translation presupposes; above the lower rungs, "translation" is partly a category error. Second, an *evaluation crisis*: headline accuracies are unfalsifiable as translation metrics — no ground-truth intended message exists to score against. The lower rungs resemble heavier-than-air flight—hard but grounded, with existence proofs in hand—while open-ended translation of a non-linguistic signal resembles alchemy, whose target may not exist. The survey ships the instrument its diagnosis calls for: the Canine Ladder Benchmark, which fixes for every rung the error signal, the metric, and the baseline a system must beat, and a ten-item checklist applied to the claims the field has already made in public. We close with a capability roadmap, distance estimates, and an ethics framework that keeps the animal's welfare as the terminal criterion. Validated affect-and-intent inference is plausible within this decade: the achievable goal, and worth building.

CCS Concepts:

- • Computing methodologies → Artificial intelligence → Natural language processing
- • Computing methodologies → Machine learning
- • Human-centered computing → Human computer interface (HCI)
- • Applied computing → Life and medical sciences
- • General and reference → Cross-computing tools and techniques → Evaluation

Keywords: animal communication, bioacoustics, foundation models, canine cognition, DogFACS, semantic grounding, interspecies interaction, evaluation, research ethics.

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1 Introduction: The Oldest Silence

For at least fifteen thousand years humans have lived alongside dogs [195], and for nearly as long we have talked *at* them without ever talking *with* them. Folktales grant the hero a magic fern-seed or a drop of dragon's blood that lets him understand the speech of beasts; the wish is older than science. What is new is the scale of the longing, and the arrival of a technology that claims to satisfy it.

Roughly **900 million dogs** live on Earth [221], some **half a billion** of them as companions in human homes [222]. About **65 million U.S. households** own a dog [224, 221], a quarter of European households keep one [223], and tens of millions more do so across Asia [226]. These households do not treat their dogs as mute appliances: in one national survey **58% of owners report regularly conversing** with their pet, precisely because the animal is experienced as a family member [225]. The picture is striking in its quiet way: hundreds of millions of people hold daily, affectionate, one-sided conversations with a companion who cannot reply in any shared language. That asymmetry, intimate cohabitation without mutual intelligibility, is the regret this survey takes as its starting point.

The old wish now arrives packaged as a product. A smartphone application records a bark and returns a sentence, "I want to play," "my tummy hurts," "I love you," in under a second, with a large advertised accuracy. The curious reader is left with two questions that turn out to be surprisingly hard to answer: *how far are we, actually, from translating what a dog says?* and, more fundamentally, *is the thing possible at all?*

We refuse to answer with a single number. "Translation" is not one capability but a stack of them, and the distance to each layer is different in kind. A system that reports whether a dog is relaxed or distressed is doing something real and useful; a system that claims to render the dog's "words" as an English sentence is doing something else. Our central move is to make that stack explicit, a five-rung *translation ladder* (Section 5), and to locate current research and current products on it. The headline finding is easy to state and, we will argue, well supported: **today's best systems live on the second rung, affect classification, with a handful of careful laboratory studies reaching toward the third.** Everything above that remains either aspirational or, in the strong sense, ill-posed.

Four currents make this the right moment for such a survey. The human translation stack has matured into a reusable toolkit, from sequence-to-sequence learning and attention to the Transformer, large-scale pretraining, multilingual translation, and speech/audio foundation models [39, 40, 41, 44, 79, 82, 62, 63, 114, 109, 137] (Section 2). Bioacoustic foundation models have matured in their own right and now transfer across species on standardized benchmarks [24, 25, 26, 27]. Flagship decoding projects have produced concrete, peer-reviewed structure discoveries in sperm whales, dolphins, and songbirds [19, 21, 29]. And canine cognition has its own empirical renaissance, from the ELTE Budapest group's bark and growl psychophysics to the UC San Diego soundboard studies [1, 4, 15, 16]. The intersection of these four currents is the space this paper maps.

A survey that only estimates distances implicitly assumes the destination exists, so we also adjudicate feasibility against three historical templates (Section 7). *Alchemy* pursued a target that rested on a category error; no amount of effort would reach it. *Heavier-than-air flight* was widely declared impossible by competent authorities, yet rested on sound physics and fell once the right mechanisms were assembled. *Perpetual motion* is provably impossible, ruled out by conservation laws. Our layered verdict: the lower rungs (affect, intent) resemble flight, hard but grounded with existence proofs already in hand; the upper rung (open-ended sentence translation of a non-linguistic signal) resembles alchemy, because the target may not exist. Distinguishing these two is the survey's most consequential act.

The thesis, in one paragraph. A dog's communicative repertoire is rich, multimodal, and

genuinely informative, but it is, in the technical sense, a *signaling system* rather than a *language*. It encodes affect, arousal, motivational state, individual identity, and some referential content, yet it shows no evidence of the open-ended, recursive, generative syntax that human language translation presupposes [187, 189]. Consequently, "translating what a dog says" is, above the level of affect and intent, partly a *category error*: there may be no sentence-shaped meaning in the signal to recover. This reframes the achievable goal, validated and calibrated affect-and-intent inference, and exposes why the field's most quotable numbers ("81.5% accuracy," "94.6% accuracy") are, as presently defined, close to meaningless: there is no ground-truth intended message to score a translation against. We call this the *falsifiability problem*, and it is the methodological heart of the survey (Section 8).

The paper is written from a computer-science and AI standpoint, with biology, linguistics, and ethics as supporting witnesses. It contributes a tech-stack synthesis and transfer audit (Sections 2–3), a feasibility adjudication and the ladder (Sections 7 and 5), a critical audit of commercial "pet translators" and the evaluation crisis (Section 8), a falsifiable benchmark protocol with a reporting checklist that turns the audit into something a lab or a vendor can be scored against (Section 9), and an honest distance estimate with a research agenda and ethics framework (Sections 10–11). Throughout, we keep two registers separate: what is *measured* (peer-reviewed effect sizes, model metrics on defined tasks) and what is *claimed* (vendor accuracies, press-release superlatives). The gap between those registers is, in a real sense, the subject of this paper.

1.1 How this survey differs

Each of the four currents has its own review literature: transfer learning has two standard treatments and a comprehensive update [146, 147], generation-side reliability has a hallucination survey [157], canine behaviour and cognition are covered by a reference monograph [252], and the soundboard controversy now has its own review [17]. None of them places a consumer claim, a Transformer paper, a bark-psychophysics experiment and a whale-coda model on one scale. The contribution here is the scale itself: an ordering of capabilities by the error signal each one requires, which lets four literatures be read as a single argument and makes the position of every published system and every marketed product a checkable statement rather than a matter of taste. Table 1 puts the neighbouring reviews beside that claim, because a survey is entitled to be read as a delta: what each of these already supplies, and what is left open until now.

1.2 Method and reference tiers

Sources were gathered in November 2026 through structured search of scholarly indexes (Google Scholar, Semantic Scholar), preprint servers, publisher sites, and project reports, using query families derived from the ladder and the tech stack; the documented search log (`sources/search-log.md`) records queries and dates, and `sources/corpus_stats.py` regenerates every count in Section 1.2 and Figure 1 from the bibliography file itself, writing a per-entry CSV (`sources/corpus_stats.csv`) that assigns each reference to a year and a thematic group. Inclusion required a direct bearing on at least one rung, one pipeline stage, or one transfer lesson; vendor marketing without a methodological claim was excluded except as an object of audit. Every reference falls into one of four tiers: **T1** peer-reviewed venue papers; **T2** preprints and technical reports, labeled as provisional; **T3** vendor claims, reported *as claims* and never as results; **T4** journalism and industry statistics, used for scale and context only. The tier, not the count, determines how much load a citation may bear. This is a narrative survey, not a PRISMA systematic review; entries confirmed only via secondary indexes should be checked against the primary record before reuse.

Table 1: This survey against the review literature it draws on. "Covers" and "Leaves open" are our assessment of each work's stated scope, not a verdict on its quality; every work listed is cited in the body.

Work	Discipline / year	Covers	Leaves open
Miklósi [252]	ethology monograph, 2007	dog behaviour, evolution and cognition as a whole field	vocal signalling as one channel among many; no decoding model, no capability ordering
Udell et al. [253]	*Biol. Rev.*, 2010	dogs' sensitivity to human actions and its domestication history	the reverse channel, from dog to human, and any measurement of it
"Talking dogs" review [17]	*Anim. Cogn.*, 2025	soundboard training and test claims in companion animals	natural signals; the machine-translation stack; consumer products
Bastos & Rossano [18]	*Interaction Stud.*, 2023	soundboard use across species, and what a symbol counts as	rung placement of the resulting evidence
Stowell [171]	*PeerJ*, 2022	deep learning in bioacoustics, with a research roadmap	canine targets; affect and intent labels; claims made to the public
Transfer-learning surveys [146, 147]	IEEE TKDE / *Proc. IEEE*, 2010–21	what transfers between distributions, and when it does not	whether any of it transfers to a non-linguistic signal
Hallucination survey [157]	*ACM Comput. Surv.*, 2023	generated text that is fluent and unfaithful to its source	the case where the source contains no message to be faithful to
This survey	CS/AI, 2026	one error-signal-ordered scale over all four currents, a benchmark, and a checklist applied to the field	the rungs themselves: Rung 4 is unpopulated and Rung 5 is not certifiable

1.3 The shape of the corpus

Figure 1 adds up the 257 entries, and the shape carries a message of its own. Seven in ten arrive after 2015; the 2020–2023 block alone holds 89, which is the foundation-model era in one breath — self-supervised audio encoders, whale and dolphin codas, the soundboard dogs, and the first consumer translators all land there. The thematic split is more diagnostic still. The human language-technology stack (machine translation and the Transformer 36, speech, audio and neural codecs 39, large language models and multimodality 32: 107 entries between them) outweighs everything canine (48) by more than two to one, and the largest single block is canine communication itself, closely followed by the methods that would have to transfer to it (transfer learning and evaluation, 40) and by cross-species decoding (35), because that is where these methods have been tested on non-human signals at all. This survey is accordingly engineered rather than ethologically centred: the tools come from human language technology, and the dog supplies the signal.

2 The Human Translation Stack and the Acoustic Toolkit

Before asking what machines can make of a bark, we should be precise about what machines already make of human speech and text. The last decade of machine translation is not a single invention but a layered stack, and each layer supplies a tool that animal decoding either inherits or lacks. Figure 2 gives the timeline; Tables 2 and 3 condense the two halves of the stack.

Statistical MT treated translation as noisy-channel decoding over phrase tables learned from parallel corpora such as Europarl [53] and OPUS [54], toolkitted by Moses and, later, by fairseq and the Transformers library [52, 245, 246]. The neural turn replaced hand-built phrases with

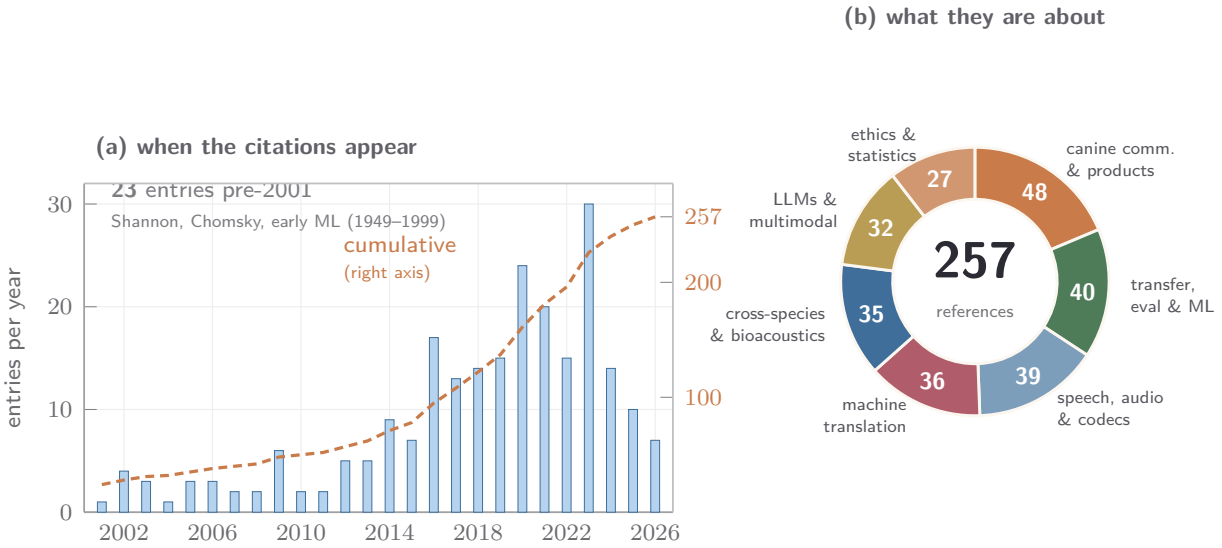


Figure 1: The corpus behind this survey. (a) Bibliography entries per year, taking the latest year stated in each entry; the 23 entries predating 2001 (Shannon, Chomsky, early machine learning) are counted in the cumulative line but not drawn as bars, and 2026 is partial. (b) The same 257 entries in seven thematic groups: canine communication, cognition and products, 48; transfer, evaluation and ML methods, 40; speech, audio self-supervision and neural codecs, 39; machine translation and the Transformer, 36; cross-species decoding and bioacoustics, 35; large language models and multimodality, 32; enrichment, ethics and population statistics, 27. Both panels are produced by `sources/corpus_stats.py`, which reads the bibliography itself, so every number in the figure is auditable entry by entry and regenerates when the bibliography changes.

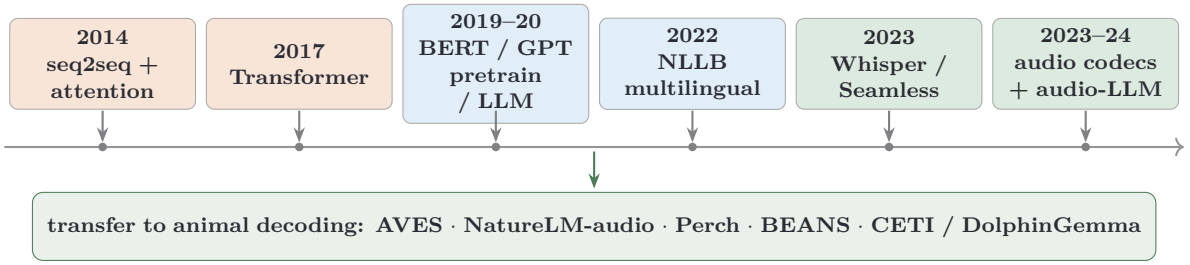


Figure 2: The human translation / Transformer genealogy, condensed. Each layer supplies a tool that animal decoding inherits (acoustic encoding, discretization, sequence modeling, multilingual transfer) or lacks (a reference corpus, shared grounding). The lower band marks where the toolkit has already been pointed at animals.

end-to-end encoder–decoder learning [39, 40], and the decisive ingredient was attention, a soft learned alignment between source and target positions [41, 42]. Convolutional variants competed briefly [43], and production systems showed that deep recurrent attention models could be deployed at scale [45, 48]; systematic architecture exploration later confirmed which design choices actually matter [244]. Two data-side inventions proved as important as the architecture: subword segmentation (BPE), which composes rare words from frequent pieces [50], and back-translation, which manufactures parallel signal from monolingual text [51, 49]. Both speak directly to the animal case: BPE is a recipe for discretizing a continuous signal into reusable units, back-translation a recipe for learning when no parallel corpus exists.

The Transformer dispensed with recurrence entirely [44], and its real legacy is that it made pretraining scalable: masked language modeling (BERT) [79], causal modeling (GPT) [80, 81, 82], and seq2seq denoising (T5, BART) [83, 84] turned massive unlabeled corpora into general-purpose representations, with refinements (RoBERTa, XLNet, ELECTRA, DeBERTa) establishing

that the objective and data regime matter more than architectural cleverness [85, 86, 87, 88]. Contextual embeddings had already shown the value of pretraining over static vectors (word2vec, GloVe, ELMo, ULMFiT) [75, 76, 77, 78]. Scaling laws make returns predictable [101], compute-optimal training clarifies the data/scale trade-off [102], and the foundation-model paradigm frames both opportunity and risk [100]. For bioacoustics the lesson is that a large unlabeled corpus of vocalizations, pretrained with a masked or contrastive objective, is the right starting point.

Human MT's most encouraging result for animal decoding is that quality does not require a bespoke parallel corpus per language pair. Shared-encoder multilingual models enable zero-shot transfer [46, 47]; cross-lingual pretraining (XLM, XLM-R, mBART, multilingual BERT) shares structure across languages [91, 89, 74, 90]; mined parallelism (WikiMatrix, CCAligned) plus many-to-many training (M2M-100, NLLB) brought hundreds of low-resource languages into scope [67, 68, 64, 62, 63]; and fully unsupervised MT is possible from monolingual corpora alone [65, 66]. This is the closest human analogue to the animal situation: a target with no parallel corpus, where structure must be bootstrapped from raw signal and transferred from better-resourced relatives.

Translation then moved end-to-end to speech: listen-and-translate proofs of concept [69], direct speech translation [70], direct speech-to-speech models [71], and unified multilingual multimodal systems (SeamlessM4T, Seamless) [72, 73]. The acoustic-encoder-plus-semantic-decoder template is exactly what a genuine animal translator would need, and it is engineerable when the source is human language. What animals lack is the evaluation layer: human MT rests on accumulated reference translations, BLEU [55], METEOR [56], and learned metrics such as COMET [57], validated against expert judgment [58] and adjudicated annually at WMT [59, 60, 61]. Every one of these presupposes a ground-truth target sentence.

Open and proprietary LLMs (LLaMA, Llama 2, PaLM, GPT-4, Gemini, Claude, Gemma) [92, 93, 247, 94, 95, 96, 97], instruction tuning [98], and chain-of-thought reasoning [99] supply the semantic and interactive layer; whether these capabilities are genuinely emergent or an artifact of the metrics we choose is itself contested [248, 249]. Vision-language models (CLIP, Flamingo, LLaVA, BLIP-2) [103, 104, 105, 106] show how to bind a non-linguistic modality to language, the exact operation a dog translator attempts when grounding a bark to a context.

A bark is not human speech, so the acoustic front-end matters. The decisive idea is to learn a general representation from raw unlabeled waveform and fine-tune on small labeled tasks: CPC [107], wav2vec [108], wav2vec 2.0 [109], HuBERT [110], WavLM [111], and data2vec [112, 113]. Two extensions matter directly for animals: wav2vec-U learns recognition without any transcriptions [126], and multilingual scaling (XLS-R, MMS, USM) pushes one encoder across thousands of languages [127, 128, 129]. The transcription-free recipe is the closest human precedent for a target with no parallel corpus.

Recognition architectures solve the alignment problem in two regimes: CTC marginalizes over all alignments between frames and labels [115, 116, 117], while attention-based listen-attend-spell learns a soft alignment [118]; the Conformer marries convolution to self-attention [119]. For animal signals the problem is the same in form but harder in substance, because there is no canonical label sequence to align to. None of this would exist without shared data and evaluation: LibriSpeech [120], Common Voice [121], Multilingual LibriSpeech [122], TIMIT and Switchboard [123, 124], and benchmarks such as SUPERB [125] and CoVoST [130, 131]. The absence of a "LibriSpeech of barks" is a concrete, fixable bottleneck.

A second lineage turns waveform into discrete tokens a language model can predict: WaveNet [132], VQ-VAE [133], Jukebox [134], SoundStream and EnCodec [135, 136], AudioLM [137] and its music and sound successors [138, 139, 140], VALL-E [141], UniAudio [142], neural text-to-speech (FastSpeech, HiFi-GAN) [251, 250], and audio-LLMs (AudioPaLM, SALMONN, Qwen-Audio) [143, 144, 145]. This is the discretization step a "bark alphabet" would require, the strategy Project CETI applied to whale codas [19].

The same representation revolution in vision, AlexNet and ImageNet [218, 219, 220], ResNet [205], and the Vision Transformer [206], is the lineage DogFACS-style facial analysis rides on (Section 4). And the toolkit has been pointed at animals: BirdNET [170], the reviews charting computational bioacoustics [171, 172], AVES [24], NatureLM-audio [25], Perch [26], BEANS [27], and OpenBEATs [28]. The transfer is measured, not assumed: fine-tuned animal-audio encoders report gains over from-scratch baselines on BEANS and BirdSet. What none supplies is a reference translation; they classify, embed, and retrieve, but do not render a bark as a sentence with measured fidelity. The engineering of the front-end is largely solved; the missing piece is semantic, not acoustic.

Table 2: The human translation / Transformer genealogy and its transfer value to animal decoding. "Transfer value" is our assessment.

Year	Milestone	Key mechanism	Transfer value to animals
2006	CTC [115]	alignment-free labeling	label variable-length vocalizations
2014	seq2seq [39, 40]	encoder–decoder	signal sequence → symbol sequence
2015	attention [41]	soft alignment	align bark segments to labels
2016	BPE / back-translation [50, 51]	subwords; synthetic parallelism	discretize; bootstrap without parallel data
2017	Transformer [44]	self-attention	the universal backbone
2017	multilingual NMT [46]	shared encoder, zero-shot	one encoder across species
2019–20	BERT / GPT / T5 [79, 82, 83]	masked / causal pre-training	foundation encoder from raw signal
2021	ViT / CLIP [206, 103]	cross-modal transformers	bind non-linguistic modality to language
2022	NLLB / Chinchilla [62, 102]	mined parallel; compute-optimal	low-resource recipe; data-vs-scale trade-off
2023	LLMs / Whisper / Seamless [94, 114, 72]	scale; weak-supervision ASR	semantic layer; robust acoustic encoder
2024	NLLB (Nature) [63]	200 languages deployed	proof that low-resource scale works

Table 3: The speech/audio foundation-model toolkit and its transfer value. SSL = self-supervised; codec = discrete acoustic tokens.

Model / family	Type	Transfer value to animals
wav2vec 2.0 / HuBERT / data2vec [109, 110, 112]	SSL	foundation bark encoder from unlabeled audio
wav2vec-U [126]	SSL	learn without a parallel corpus
XLS-R / MMS / USM [127, 128, 129]	SSL	one encoder across species
CTC / LAS / Conformer [115, 118, 119]	ASR	map variable-length signal to labels
Whisper [114]	ASR	robust off-the-shelf acoustic encoder
LibriSpeech / Common Voice / SUPERB [120, 121, 125]	data	template for a "bark corpus"
SoundStream / EnCodec [135, 136]	codec	discretize vocalizations into tokens
AudioLM / VALL-E / UniAudio [137, 141, 142]	codec-LM	search for sequence structure
AudioPaLM / SALMONN / Qwen-Audio [143, 144, 145]	audio-LLM	bind acoustic signal to language
AVES / NatureLM / Perch / BEANS [24, 25, 26, 27]	bioacoustic	cross-species transfer, already realized

3 Transfer Audit and Seven Lessons

With the two stacks in view, the decisive question is which of their *presuppositions* survive when the source is a dog rather than a human. Transfer learning formalizes when source knowledge helps a target domain [146, 147], and deep features are often transferable, with generality decreasing from early to late layers [148]. This is the optimistic half: an encoder pretrained on human speech or general audio can be fine-tuned on barks, and cross-species models already demonstrate the gain [24, 26]. The machinery underneath is modality-agnostic: backpropagation [216], LSTM [217], the deep-learning synthesis [213, 214, 215], and the optimization and regularization practices that make large models trainable (Adam, dropout) [203, 204]. Nothing in these assumes human language, so a bark encoder built from them is engineerable. But transfer of features is not transfer of semantics: the source domain of human MT carries a target language with shared reference; the animal target domain carries no such thing. Feature transfer succeeds precisely where the task is classification; it cannot manufacture a reference translation the source never had.

The break is sharpest at grounding. Human translation is grounded because source and target symbols refer to a shared world of objects, actions, and states [159]; even text-only models inherit this indirectly through human usage, and the limits of that inheritance are well documented [160]. A dog's signal is grounded in *its* affective and motivational states, not in a public lexicon, so mapping it to an English sentence is interpretation across a referential gap, not translation between two grounded languages. Compositionality raises a second wall: whether neural models learn it is contested [161], diagnostic benchmarks show standard architectures often fail to generalize compositionally [163, 162], and probing reveals that BERT-style representations encode syntax without necessarily encoding meaning [164, 165]. Even if a bark sequence had compositional structure, our models might not recover it; and the likely absence of compositionality in the signal means the very property that makes human MT work is missing at the source.

Two further lineages matter for validation rather than decoding. Generative models (GANs, diffusion, latent diffusion) [207, 208, 209] could synthesize candidate vocalizations for playback, and reinforcement learning (DQN, PPO, self-play) [211, 210, 212] supplies the framework for an interactive agent that learns which inferred intents predict the dog's behavior. These transfer well; what they require is an error signal, which the field currently lacks. Table 4 summarizes the audit: the acoustic and representational layers transfer almost intact, the semantic and evaluative layers do not transfer at all. Where a row says "transfers," the claim is backed by measured cross-species fine-tuning gains; where it says "fails," it rests on the absence of any reference corpus or shared lexicon.

Table 4: Transfer-feasibility audit: what each layer of the human stack presupposes, and whether that presupposition holds for animal signals.

Component	Human MT presupposes	Animal reality	Verdict
Acoustic encoder	raw waveform, any language	raw waveform, any species	transfers
Self-supervision	large unlabeled corpus	large unlabeled barks exist	transfers
Subword / codec units	discrete reusable units	units discoverable (cf. CETI)	partial
Sequence model	recursive, compositional source	likely non-compositional signal	limited
Parallel corpus	bilingual reference text	no reference exists	fails
Grounding	shared public lexicon	private affective states	fails
Evaluation metric	ground-truth target sentence	no intended message to score	fails
Closed-loop validation	observable error signal	buildable via playback / RL	buildable

Read as a computer-science narrative, this history distills into seven lessons for anyone attempting to decode an animal (Table 5). The most robust finding of the last decade is that a large unla-

beled corpus pretrained with a masked or contrastive objective beats a clever architecture trained from scratch [79, 109, 85], so the first investment is a public bark corpus and a self-supervised encoder, not a new model. Because animals have no parallel corpus and never will, the supervision must be bootstrapped the way human MT learned to manufacture its own [51, 49, 65, 66]: generate candidate vocalizations, play them back, and let the animal's response supply the error signal. And because scaling laws make returns predictable [101, 102], they tempt the inference that a big enough model will eventually translate a bark; the grounding problem is a referential mismatch, not a data deficit, so scale helps Rungs 1–3 and cannot fix a category error at Rung 5. Finally, modern systems render fluent sentences with a language model [82, 98], and fluency is easily mistaken for fidelity, exactly the failure the hallucination and faithfulness literature documents [157, 158]. The rendered sentence is a user-interface layer; the honest number lives upstream, at the classifier.

Table 5: Seven lessons from human machine translation, each paired with a prescription for animal decoding.

#	**Human-MT finding**	**Prescription for animals**
1	pretraining beats architecture [79, 109]	build a large unlabeled bark corpus + SSL encoder first
2	subwords / codecs unlock generalization [50, 19]	discover reusable bark units before claiming translation
3	unsupervised / back-translation [65, 51]	bootstrap supervision via playback + RL, not annotators
4	multilingual sharing helps low-resource [62, 46]	one shared canid encoder across species and individuals
5	references + shared tasks = credibility [55, 59]	no error signal, no "translator"; disclose protocols
6	scaling laws predict returns [101]	scale helps Rungs 1–3; grounding, not scale, blocks Rung 5
7	fluency $\neq$ fidelity [157, 158]	label the sentence as interface; score the classifier

4 What Dogs Actually Signal

Before asking whether a machine can translate a dog, we must ask what there is to translate. Canine communication is multimodal, graded, and heavily context-dependent, and it is asymmetric: the dog's receptive understanding of human signals far exceeds its productive repertoire. Table 6 summarizes the channels.

The domestic dog is unusually vocal for a canid, a trait widely interpreted as a product of domestication [252]. The ELTE Budapest ethology group built the most systematic evidence on what barks encode: human listeners classify barks recorded in different situations above chance [1], acoustic parameters (fundamental frequency, tonality, inter-bark interval) carry emotional information listeners reliably extract [2], and barks carry individual identity, with machine classifiers recovering context and individual from kennel recordings [3]. Growls are even more informative: listeners judge a dog's size and aggressive intent from growl formants, and food-guarding versus play growls are acoustically and perceptually distinct [4]. Whines and whimpers complete a repertoire that is graded rather than discrete. The engineering literature has begun to automate exactly these judgments: the ICASSP 2018 study framed bark analysis as automatic recognition of context and perceived emotion [5]. The nuance that matters for Section 8 is that the target label is the human-perceived emotion or the eliciting context, not the dog's "intended message."

Canine facial behavior is codable with the same rigor as human facial action coding. DogFACS adapts Ekman and Friesen's FACS to the dog face [6], and using it Kaminski et al. showed that dogs produce markedly more facial movement, especially the inner brow raiser that enlarges the

apparent eye, when a human is attending to them than when the human's back is turned, with food playing no role [7]. The "puppy dog eyes" expression thus looks like a communicative signal tuned to human attention, possibly shaped by domestication; a 2024 comparative study further reported that domestication constrains facial expressiveness relative to wolves [8]. Automated facial analysis is now an active vision subfield, with explainable CNNs, improved optimization, and end-to-end expression and landmark pipelines [9, 10, 11, 12]. These are genuine advances, but their labels are human-coded affective categories or action units, not propositions.

A large fraction of everyday signaling is postural: ear position, tail carriage and wag direction, weight distribution, piloerection, gaze. Tail-wagging lateralizes by emotional valence [36], and the ethogram of dominance, submission, play, and stress is well documented in the clinical literature. Body signals are often more diagnostic than vocalizations, a point vendors themselves concede when they fuse body language and face to reach claimed accuracies. Olfactory signaling is enormously important to dogs but essentially inaccessible to consumer sensors; we flag it as a blind spot, not a solved channel.

The receptive side is impressive. Rico the border collie provided the first evidence of "fast mapping" in *Science* [13], and Chaser was documented retrieving 1,022 named objects and responding to simple noun–verb combinations [14]. These are exceptional, intensively trained individuals, but they establish an upper bound on referential comprehension far above the productive repertoire. A future "dialogue" system may therefore succeed by letting dogs receive structured human input long before it can read structured canine output.

The most direct attempt to give dogs a productive symbolic channel is the augmentative interspecies communication soundboard, a grid of buttons that speak recorded words when pressed, now under controlled study by the UC San Diego *TheyCanTalk* project. Two 2024 peer-reviewed papers anchor the science: soundboard-trained dogs respond appropriately to words such as "play" and "outside" whether spoken by the owner or triggered by an unfamiliar button-presser, indicating word comprehension rather than cue-reading [15]; and two-button combinations are non-accidental, non-random, and non-imitative, evidence of intentional if linear combination [16]. A 2025 review synthesizes the evidence and the skepticism [17]. The consensus interpretation is careful: the combinations look like associative sequences (name + action, place + go), not recursive grammar; spatial and immediate concepts are robust while temporal abstractions are rare and inconsistent. Soundboards demonstrate referential intent on the third rung, but by giving the dog a human-designed symbol set, not by decoding its natural signal.

The substrate is real and rich: dogs signal affect, arousal, identity, motivation, and, with human scaffolding, referential intent. What it does not contain is an open-ended generative grammar. This single fact, more than any limitation of current models, bounds what "translation" can mean.

5 The Translation Ladder

The phrase "translate what a dog says" smuggles in an assumption: that a sentence-shaped meaning waits in the signal, the way a French sentence waits behind its English translation. The biological record gives reason to doubt this. We therefore replace the single verb with a ladder of five distinct capabilities, each with its own input–output contract, its own notion of ground truth, and its own distance from the present (Figure 3).

Rung 1, signal detection and segmentation, asks only whether there is a vocalization, where it begins and ends, and who produced it. This is a classical detection problem with objective ground truth, and it is essentially solved: bioacoustic detectors segment and attribute calls reliably [26, 24], and individual identification from bark acoustics is above chance [3]. Nothing about Rung 1 requires understanding.

Rung 2, affect and arousal classification, maps a signal to a small set of affective categories

Table 6: Canine communication channels, the information they carry, and the strongest evidence. "Decodable today" means validated above-chance inference on a defined label, not sentence-level translation.

Channel	Information encoded	Key evidence	Decodable today
Bark	affect, arousal, context, identity	Pongrácz et al. [1, 2]; Molnár et al. [3]	affect/context
Growl	size, aggressive intent, guarding vs. play	Faragó et al. [4]	size/intent
Whine	separation distress, submission	ethological observation	coarse affect
Face (DogFACS)	attention-directed signals, affect	Waller et al. [6]; Kaminski et al. [7]	action units; affect
Body/tail/gaze	motivation, stress, play, social intent	ethograms; wag laterality [36]	posture classes
Olfaction	identity, reproductive state, territory	scent-marking literature	not sensor-accessible
Reception	human word referents (up to 1,022)	Kaminski et al. [13]; Pilley & Reid [14]	comprehension, not production
Soundboard	referential intent via human symbols	Bastos et al. [15, 16]	word-level intent (trained)

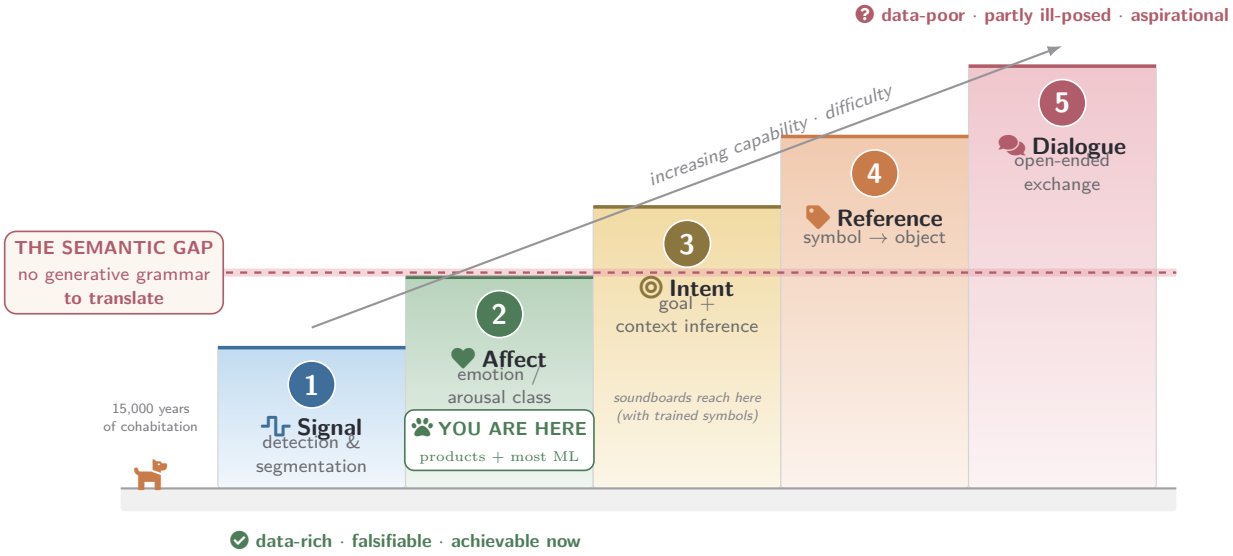


Figure 3: The translation ladder. Current consumer products and most machine-learning work occupy **Rung 2** (affect classification); controlled soundboard studies reach **Rung 3** (referential intent) but only by supplying a human-designed symbol set. The dashed boundary marks the *semantic gap*: above it, the dog's natural signal lacks the recursive, generative syntax that sentence-level translation presupposes, so the target becomes possibly ill-posed rather than merely hard.

or to continuous arousal/valence dimensions. This is where nearly all commercial "translators" live and where the peer-reviewed ML literature is strongest, because the labels are human-perceived emotions or eliciting contexts that annotators can agree on [2, 5, 9]. Rung 2 is falsifiable and genuinely useful for welfare monitoring. Its limit is that an emotion label is not a sentence; rendering "fearful" as "I am scared of the vacuum" adds narrative the signal never contained.

Rung 3, intent and context inference, asks what the dog is trying to do, given the surrounding context, and requires multimodal fusion and a model of the dog's goals. Soundboard studies provide the cleanest evidence, because the button press makes the intent explicit and checkable [15, 16], but the symbol set is human-designed and trained. Rung 3 on natural signals

is far less developed and much harder to validate.

Rung 4, referential semantics, would map a spontaneous canine signal to a specific referent without human-supplied symbols. Receptive studies show dogs can understand such mappings [13, 14], but a productive, open referential vocabulary in natural dog communication has not been demonstrated. This is the first rung where the translation framing strains: there may be no stable symbol-to-referent dictionary to recover.

Rung 5, open-ended dialogue, is the folkloric dream: free, generative, two-way conversation. It presupposes a recursive generative grammar on the canine side, and the evidence of graded affective signals, linear associative combinations, and no demonstrated syntax gives no support for that presupposition. Rung 5 is therefore not merely distant; in its strong form it is possibly ill-posed. A weaker engineered version, structured human-to-dog input via soundboards with the dog responding through trained symbols, is conceivable and already nascent.

The ladder converts a vague question into answerable ones: which rung, with what ground truth, at what validated accuracy? It also explains the rhetoric of the market, where vendors demonstrate Rung 2 while marketing Rung 5. The rest of the survey uses it as a ruler.

6 The Pipeline and the Cross-Species Evidence

If genuine translation were possible, what would the system look like? We decompose it into four stages, encoding, structure discovery, semantic grounding, and interactive validation, and read the flagship cross-species projects as evidence for each. The stages are a loop rather than a waterfall: validation feeds back into grounding, which constrains structure, which shapes the encoder. Table 7 places the milestones on a timeline.

Encoding is the mature front. AVES adapts the HuBERT/wav2vec paradigm to animal sound [24]; NatureLM-audio extends this to an audio–language foundation model that detects, captions, and generates animal sounds [25]; Perch shows a single encoder can serve many taxa [26]; and BEANS, BirdSet, and OpenBEATs give these encoders a common, measured evaluation surface [27, 28]. For dogs, a strong Rung 1–2 encoder is available off the shelf; the bottleneck is labels and semantics, not representation quality.

Structure discovery is where the flagship projects have produced their most cited results. Project CETI's analysis of 8,719 sperm-whale codas revealed a "phonetic alphabet": rhythm, tempo, rubato, and ornamentation combine contextually, expanding the inventory of distinguishable forms nearly tenfold [19]. The 2022 program paper lays out the ambition [20], the 2025 WhAM model learns to complete and generate whale-like codas [22], and a late-2025 preprint proposes a translative model of whale vocalization [23]. Yet the investigators are careful: the alphabet describes how signals are *built*, not what they *mean*. Google's DolphinGemma takes the language-model route directly, a roughly 400-million-parameter Gemma-based model with a SoundStream tokenizer, trained on four decades of Atlantic spotted dolphin recordings, that predicts the next sound as an LLM predicts the next token [21]. The Earth Species Project's 2026 zebra-finch model (ZF-AIM) goes further, detecting, predicting, and generating calls in real time and engaging birds in turn-taking exchanges [29]. The lesson for dogs is double-edged: the methods are species-agnostic, but the structure they find is statistical, and predicting the next sound is not understanding it. Yovel and others warn that such models may cluster calls by acoustic similarity while missing meaning entirely [30]. For a repertoire as graded and context-bound as the dog's, the prior probability of a clean combinatorial alphabet is lower than for whale codas or birdsong.

Grounding ties a signal to the world: the eliciting context, the subsequent behavior, the objects in view, the social partner. Dogs are paradoxically advantaged here, because they live in richly instrumented human homes and their behavior is legible to us. The soundboard is effectively a grounding device: the button press, the spoken word, and the consequent action form a triangu-

lated, checkable mapping [15, 16]. The Wild Dolphin Project's CHAT system does the analogous thing underwater, pairing a synthetic whistle with a coveted object so the dolphin can request it by name [21]. The general principle, ground symbols through interaction and consequence rather than passive correlation, is the field's most important methodological insight, and exactly what consumer bark-to-sentence apps lack.

Validation closes the loop: does the animal treat the model's output as meaningful? The gold standard is the playback experiment, used for decades to establish referential alarm calls, with ESP's interactive finch work and CHAT as modern model-driven versions. For dogs, the loop would infer an intent, emit a candidate signal or action, and test whether the dog's behavior confirms the inference, with the experimenter blind to the model's guess to avoid Clever-Hans contamination. No consumer product does this; it is the dividing line between science and entertainment.

Reading the cross-species literature through the ladder, three things transfer to dogs: the self-supervised encoders, the structure-discovery toolkit, and above all the interaction-grounding and validation protocol. Three things do not: the assumption of discrete culturally transmitted units, the luxury of decades of labeled wild corpora, and the clean playback accessibility of whales and birds. Dogs are domestic, multimodal, and graded; their advantage is legibility and proximity, their disadvantage the absence of a syntax to decode. The maturity pattern is the technical signature of the whole field: the front of the pipeline is borrowable and relatively mature, the back is where canine-specific, falsifiable work is needed and where almost no commercial effort is directed.

Table 7: Selected milestones in cross-species decoding. Type: R = peer-reviewed structure result; M = generative / foundation model; I = interactive grounding protocol; P = consumer product. These advance the toolbox; none yet delivers validated sentence-level translation.

Year	Milestone	Type	Significance
2002	Bow-Lingual [32]	P	six-emotion bark matcher; the consumer archetype
2005	ELTE bark psychophysics [1]	R	humans classify bark context above chance
2013	DogFACS [6]	R	objective facial-action coding for dogs
2018	ESP founded; ICASSP bark ML [5]	M/R	open infrastructure; automated context/affect
2022	CETI program paper [20]	R	roadmap for whale decoding
2023	AVES; BEANS [24, 27]	M	animal-sound SSL encoder + benchmark
2024	whale phonetic alphabet [19]	R	combinatorial coda structure (form, not meaning)
2024	soundboard studies [15, 16]	R/I	canine word comprehension + intentional combination
2024	NatureLM-audio [25]	M	audio–language foundation model
2025	DolphinGemma + CHAT [21]	M/I	sequence model + object-grounded interaction
2025	WhAM [22]	M	generative whale-coda model
2026	ZF-AIM finch interaction [29]	I	real-time turn-taking with a songbird

7 Feasibility: Alchemy, Flight, or Perpetual Motion

The templates introduced in Section 1 can now be adjudicated against evidence. The strongest argument that decoding animal signals is flight rather than alchemy is that it has already been done, repeatedly, in other species. Vervet monkeys produce predator-specific alarm calls that elicit appropriate escape responses, and habituation–dishabituation experiments show the calls function as referential labels [175, 176]. Prairie dogs encode predator type, size, and color [181],

chicken alarm calls carry graded referential content [182], the honeybee waggle dance communicates distance and direction [177], bottlenose dolphins use signature whistles as individual labels [173, 174], and chimpanzees inform conspecifics of danger in a goal-directed way [178]. Japanese tits go further, combining calls into compositional sequences [179], and formal-semantics analyses find context-dependent, rule-governed meaning in animal calls [180]. Referential and compositional signaling exists in nature, so decoding it is an engineering problem, not a category error. The open question is whether dogs sit in this referential class; the canine evidence places them closer to the affective end.

The dog, meanwhile, is a willing and readable partner. Domestication reshaped its social cognition for life with humans [191, 195]: dogs follow human pointing and gaze better than chimpanzees [193], look back to humans at unsolvable tasks [192], show reward-system activation to praise in fMRI [194], possess voice-sensitive brain regions homologous to humans' [37], and lateralize tail-wagging by emotional valence [36]. A few dogs learn hundreds of object names [13, 14], and soundboard-trained dogs produce non-random, non-imitative two-button combinations and respond to human button presses above chance [15, 16, 18]. The dog is not a black box: it perceives human communicative intent, signals its own states readably, and in controlled settings exchanges referential tokens. Affect-and-intent decoding has empirical ground to stand on; it is flight with a runway already built.

The pessimistic half of the verdict concerns Rung 5. The comparative evidence suggests that the recursive, generative syntax human translation presupposes, the "faculty of language in the narrow sense," may be uniquely human [187, 189, 183]. On this view the language instinct is a species-typical adaptation [188] rooted in shared intentionality [184], and the cultural-evolution literature shows how iterated learning and cumulative culture shape linguistic structure into something no individual animal invents [185, 186]; even the statistical universals of human language are signatures of a generative system [190]. If a dog's repertoire lacks this structure, "translating its sentences" is alchemy-like: not hard engineering but a target that may not exist. The modality matters: this is not a perpetual-motion proof. No conservation law forbids richer animal signaling, and the compositionality results above keep the empirical question open. Rung 5 is possibly ill-posed, not provably impossible.

The layered verdict, then: Rungs 1–2 are achieved flight; Rung 3 is flight with a foothold; Rung 4 is flight in some species and an open question for dogs; Rung 5 is alchemy-like. Nothing here is perpetual motion, and the existence proofs guarantee that something real is achievable. The practical consequence is to pour effort into the grounded, falsifiable lower rungs and treat the upper rung as a hypothesis about the animal rather than a product roadmap (Table 8).

Table 8: Existence proofs and the feasibility template each supports. Nothing here is "perpetual motion" (provably impossible).

Claim	**Existence proof**	**Template**
Affect decodable (R2)	humans classify barks/growls [1, 4]; tail lateralization [36]; voice regions [37]	flight (achieved)
Intent inferable (R3)	pointing [193]; look-back [192]; soundboard combos [16]	flight (foothold)
Reference in animals (R4)	vervet & prairie alarms [175, 181]; dolphin labels [173]; bee dance [177]	flight (species-dependent)
Compositionality (R4)	tit compositional calls [179]; formal semantics [180]	flight (rare)
Dogs read human speech	fMRI reward [194]; word learning [13, 14]	flight (achieved)
Open translation (R5)	no recursive syntax in dogs; narrow faculty possibly unique [187, 189]	alchemy-like
Provably impossible?	no conservation law forbids signaling	not perpetual motion

8 Products and the Evaluation Crisis

A small industry now sells the folkloric dream as a feature. The archetype is Takara's BowLingual (2002): a collar microphone and handheld display that matched bark acoustics to six emotions and then to one of roughly 200 canned phrases [32]. It sold well, won the 2002 Ig Nobel Peace Prize, and established the template: classify affect, then decorate it with a sentence. Two decades later the template is unchanged; only the classifier has been swapped for a deep model and the phrase bank for a language model. The current flagship, Traini, analyzes sound, images, and video to infer emotion and need, reporting about 12 emotions across roughly 120 breeds with a headline accuracy of 81.5% [31]. Its founder has publicly conceded that audio alone is insufficient and that body language and face must be fused to reach the claimed accuracy, an implicit admission that the product is multimodal affect inference; the reverse direction (human speech rendered as "dog") is far weaker, on the order of 40% in independent coverage. MeowTalk applies the idea to cats [33], the Petpuls collar classifies emotional states from vocalizations [35], and in China "Meng Xiao Yi" reports a Qwen-based model trained on millions of pet voiceprints with a claimed 94.6% emotion accuracy [34]. Table 9 audits the landscape against the ladder.

Read carefully, every headline number is an emotion-classification metric, not a translation metric. "81.5%" means that on some held-out set the model's emotion label agreed with a human-assigned label 81.5% of the time. That is a real, falsifiable Rung-2 result, if the test set, label protocol, and class balance are disclosed, which consumer vendors rarely do. It says nothing about whether the rendered sentence corresponds to anything the dog intended, because no such ground truth exists. The sentence is generated downstream of the classifier by a language model optimizing for human emotional resonance, not fidelity to the animal. The product demonstrates Rung 2 and sells Rung 5.

Three incentives keep the gap open. The consumer value proposition is emotional, so a plausible affectionate sentence beats a calibrated probability vector even though the latter is the honest output. The data flywheel favors engagement over validation: every uploaded clip enriches the classifier, but no mechanism tests whether the "translation" is correct, because the dog cannot complain. And the scientific validation loop is slow and unglamorous and yields no shareable screenshot. The result is a market optimized for the appearance of translation.

The deeper problem is falsifiability. Translation in the human sense is scored against a reference; canine communication has none. There is no canonical intended message behind a bark and no bilingual informant who can confirm that "I want to play" is what the dog meant, and per Section 4 there may be no sentence-shaped intent to report. What can be scored is the upstream classifier, and reporting that as a "translation accuracy" is a category mistake. The ELTE tradition is explicit that its labels are human-perceived emotion or eliciting context [2, 5]. "This bark means 'I love you' " admits no observation that could show it false: the dog will not contradict it, and any subsequent behavior can be retrofitted to the narrative. Compare a weather forecast, scored against tomorrow's weather, or speech-to-text, scored against a transcript. The pet translator has no tomorrow and no transcript.

Two classic threats compound this. The Clever Hans effect, the horse that appeared to do arithmetic but was reading its trainer's unconscious cues, is the permanent hazard of animal-communication experiments; soundboard studies guard against it with blind protocols and unfamiliar button-pressers [15], while consumer apps used by the devoted owner in an uncontrolled home do the opposite. Anthropomorphism is the second: human observers project their categories onto the dog [9], and a language model trained on human text amplifies the projection, risking teaching owners to ignore the dog's real body signals in favor of a flattering sentence.

A defensible evaluation program therefore has four ingredients: defined labels and disclosed protocols on a public held-out corpus, the BEANS/BirdSet discipline applied to dogs [27]; calibration, so probabilities match observed frequencies and a welfare tool can threshold responsibly;

closed-loop validation of inferred intent against the dog's subsequent behavior under blind conditions; and a strict separation of the rendered sentence from the score. None of today's consumer products satisfies these. Table 10 maps the principal canine results to their actual label source: everything at Rungs 1–2 has an objective or consensus label and is falsifiable; Rung 3 is falsifiable only because a human-designed symbol set makes the intent checkable; the bark-to-sentence claim has no label at all.

This crisis is the animal-communication instance of a broader reckoning in AI about what benchmarks measure. The "stochastic parrots" critique warns that scaling text models amplifies form without grounding [150], the "climbing toward NLU" position argues that the field mistakes benchmark gains for understanding [149], and broad suites (HELM, BIG-bench, MMLU, TruthfulQA) expose how much headline scores depend on task choice and contamination [153, 154, 155, 156]. Documentation practices such as datasheets and model cards exist to make a system's limits legible [151, 152], and the hallucination literature shows that fluent generation routinely outruns its evidence [157, 158]. Table 11 consolidates the limitations this survey has identified. We propose a simple demarcation: a canine communication system should be called a translator only if its outputs are scored against an independent, animal-grounded error signal. By this criterion no current product qualifies; the soundboard paradigm qualifies at the word level; the rest is affect classification with a narrative interface. This is the difference between a measurement and a story.

Table 9: Commercial "pet translators" audited against the ladder. Accuracies are vendor-reported claims, not independently verified; all measure emotion classification (Rung 2).

Product	Modality / method	Claim	Actual rung	Reverse
Bow-Lingual (2002) [32]	bark acoustics → 6 emotions	∼200 phrases	2	no
Traini / PetGPT [31]	audio + image + video	81.5%, 12 emotions	2	weak (∼40%)
MeowTalk [33]	cat vocalization classifier	emotion labels	2	no
Petpuls [35]	collar vocalization analysis	emotion states	2	no
Meng Xiao Yi [34]	Qwen + voiceprints	94.6% (claimed)	2	no
DogTok	human command → bark	training aid	n/a	yes

Figure 4 reads the two audits together. Counting the entries of Tables 10 and 9 by the rung each one actually reaches gives a shape no single table shows: seven falsifiable canine studies cluster at rungs 1–3, five consumer products cluster at rung 2, and right of the gap the validated column is empty while the claim column holds one entry — the bark-to-sentence category, scored at the rung it advertises. Evidence and marketing here differ less in quality than in location.

9 A Protocol That Could Settle It: the Canine Ladder Benchmark

The complaint running through Sections 8–8 is methodological, not skeptical: this field reports accuracies against labels that no independent process can check. A survey that diagnoses a measurement problem owes the field a measurement, so this section specifies one. The design rule is single and severe. **Each rung may be scored only against an error signal the model cannot see and the experimenter cannot leak.** Everything below follows from that rule, and each rung gets a task, the ground truth that makes the task checkable, a primary metric, a baseline the system must beat, and the observation that would show the rung was not actually reached.

Table 10: Evidence-type map: what ground truth each canine study actually has. "Falsifiable" means an independent error signal exists.

Study	Task	Label / ground truth	Rung	Falsifiable?
Pongrácz et al. [1, 2]	bark context / emotion	human consensus on context	2	yes
Molnár et al. [3]	individual ID	dog identity (objective)	1	yes
Faragó et al. [4]	growl size / intent	human judgment + known context	2–3	yes
ICASSP 2018 [5]	context / emotion	human-perceived label	2	yes, if disclosed
DogFACS CNNs [9, 10, 11]	facial affect / AUs	human-coded FACS	2	yes
Bastos et al. (PLoS) [15]	word comprehension	blind behavioral response	3	yes (scaffolded)
Bastos et al. (Sci. Rep.) [16]	two-button combination	combination statistics	3	yes (scaffolded)
bark-to-sentence apps [31]	rendered sentence	none (no intended message)	5 (claimed)	**no**

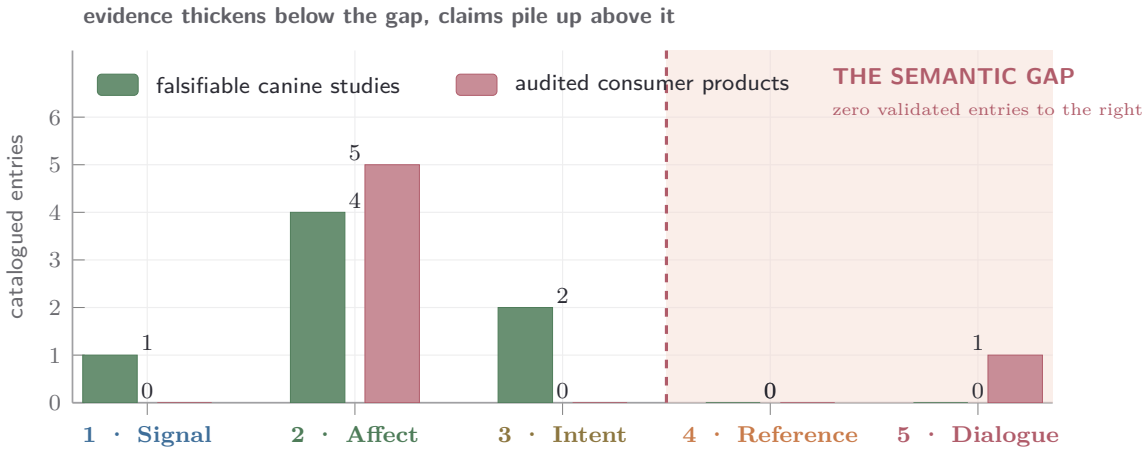


Figure 4: Catalogued entries per rung of the ladder. Green: the canine studies of Table 10 that carry an independent error signal, assigned to the rung each study reaches (the growl study spans rungs 2–3 and is counted at rung 2; the bark-to-sentence row is excluded from this series). Rose: the audited products of Table 9 at their actual rung (DogTok, a command-to-bark aid, has no receptive rung and is left out), plus the bark-to-sentence category placed at the rung it claims. The shaded region is the semantic gap of Figure 3: nothing to its right has been validated.

Table 11: Consolidated limitations of canine "translation," with locus and consequence.

Limitation	Where it bites	Consequence
No ground-truth message	bark-to-sentence scoring	fidelity is unfalsifiable
Human-perceived labels	Rung-2 metrics	a proxy for state, not a readout
Undisclosed protocols	vendor accuracies	claims cannot be checked
Clever Hans / anthropomorphism	uncontrolled home use	the observer leaks the answer
No parallel corpus	supervised training	no translation signal exists
Semantic gap / no recursion	Rung 5	the target may not exist
Fluency $\neq$ fidelity	LLM sentence rendering	hallucinated "translations"
No closed-loop error signal	consumer market	the field cannot self-correct

9.1 The five tasks

Rung 1: signal and sender. Separate vocalisation from non-vocal sound, then identify the individual. The error signal is objective: session metadata and a video-verified sender, the trick that made the individual-identification studies convincing [3]. Score equal-error rate on splits taken per dog and per recording session, so that microphone and room cannot carry identity. The baseline that must be beaten is a human speaker-verification embedding transferred without fine-tuning, because that tests whether canine vocal anatomy contributes anything a human-speaker model does not already supply [167]. A system has not reached rung 1 if its accuracy survives a session shuffle: that means it found the recording equipment, not the dog.

Rung 2: affect. Classify the animal's affective state over a pre-registered taxonomy. Human consensus alone is not ground truth here — it is a proxy, and treating it as a readout is the field's most common substitution. A scored submission needs at least two anchors fixed before any model sees the data: a manipulation the experimenter controls and records in advance (isolation, food barrier, territorial stranger, play; the design Pongrácz and Faragó used [1, 4]), and an animal-side readout that does not depend on anyone's judgement — heart-rate variability, latency to approach or withdraw, latency to settle. Human ratings are admitted as a third anchor only if collected blind to the manipulation and never used for training. Report balanced accuracy together with calibration, because an over-confident affect classifier in a consumer app is worse than a honest one that abstains [166], and report an abstention curve. The baseline that must be beaten is an arousal-only regressor built from energy and fundamental-frequency statistics. If it ties the model, the model measured arousal, not affect, and the paper must say so in those words.

Rung 3: intent and reference. Predict what the animal is acting on. The only clean error signal is the animal's own behaviour inside a forced-choice design whose alternatives are physically constrained: an object-choice task with the olfactory route eliminated, which is exactly the control that made the word - comprehension results defensible [13, 14], or a button array recorded while the handler is blind to the target. Score accuracy against a permutation baseline over the observed sequences, as the combination-analysis study did [16], and require generalisation across dogs. The rung is not reached if performance collapses when the human leaves the room; that collapse is the Clever Hans signature, and a benchmark that never tests it will keep mistaking cueing for competence. The alternative is already standard practice in animal cognition: cross-modal matching experiments score what the animal does — which of two people a horse looks at after hearing one of them speak — rather than what a human judge says the recording means [168, 169]. That swap is the difference between a measurement and a performance, and it is the single most important change this benchmark asks of canine work.

Rung 4: referential semantics, tested as composition. Map multi-element signals or sequences onto multi-element states. The error signal is a closed inventory of objects and actions with combinations held out of training, and the metric is accuracy on unseen combinations against a co-occurrence factorisation that has seen the same elements but not the same pairings. A model that matches the factorisation has learned frequencies, not structure. This is the rung where Figure 4 shows the validated column empty, so a single surviving entry here would be the most informative result the benchmark could produce in its first year.

Rung 5: open-ended dialogue. There is no protocol that can certify this rung, and the survey will not pretend otherwise: certifying it would require an independent record of an intended sentence, and no such record exists for a species without observable language. What can be built is the other half of science — a test that can reject. Run a behavioural-equivalence trial: give naive human participants a rendered sentence and a control sentence matched for affect, polarity and length, then measure what the participants subsequently do for the dog. If the two conditions are indistinguishable, the rendering carries no information beyond rung 2 and the word "translation" is refuted for that system. The test is one-sided by construction. It can close claims down; it

cannot open a rung up, and a benchmark report should state which of the two it is doing.

Table 12: The Canine Ladder Benchmark: one task per rung, with the error signal that makes it checkable and the baseline a submission must beat. The final column is the observation that shows the rung was not actually reached.

Rung	Task	Error signal	Must beat	Not reached if
1 Signal	vocal vs. non-vocal; identify the dog	session metadata, video-verified sender	transferred human speaker-verification embedding	accuracy survives a session shuffle
2 Affect	n-class state over a pre-registered taxonomy	controlled pre-recorded manipulation + animal-side physiological or latency readout	arousal-only energy/f_0 regressor	the arousal baseline ties the model
3 Intent	what the animal is acting on	forced-choice behaviour with alternatives physically constrained; handler blind	permutation baseline over sequences	performance collapses when the human leaves
4 Reference	multi-element signal → multi-element state	held-out combinations from a closed inventory	co-occurrence factorisation	factorisation matches the model
5 Dialogue	render an intended sentence	none available; use a rejection test	—	see the text: this rung is scored by falsification only

9.2 Rules that apply to every rung

Four cross-cutting requirements do most of the work, and none of them is novel — the point is that this field has not adopted them. Splits are per dog and per session, never per clip; a benchmark whose test set contains another recording of the same animal in the same room will be solved by the room. Taxonomies are pre-registered, so that a class set cannot be edited until the confusion matrix looks good. Reporting follows the artifact discipline the ML side of this survey already has vocabulary for: datasets ship with datasheets [151], models with model cards [152], and a claim about capability is written at the rung the evidence reaches, which is the discipline whose absence produced the sentences in Table 9. And abstention is legal and scored: a system that says "insufficient signal" on half of a home recording and is right about the other half is more useful than one that always guesses.

A submission is then judged on a short checklist, and the checklist is the contribution as much as the tasks are: per-dog split declared; error signal named and independent of the annotator; arousal baseline reported; human raters blind; handler absent or blinded; calibration reported; abstention reported; negative controls included; taxonomy declared before training; rung stated in the abstract. Ten items, answerable in a page.

Table 13 answers them for the four classes of system this survey has already audited, using only what those sources disclose. The audit cannot tell an absent practice from an undisclosed one, and in this literature the distinction has no consequence: a missing control and an unreported control both leave a claim unchecked. Two numbers carry the section's argument. The systems marketed as translators satisfy none of the ten items outright. The two bodies of work that score well — the symbol-array studies of animal cognition and the open bioacoustic benchmarks — are the two that were never built to sell a sentence, which is the strongest available evidence that the protocol asks for ordinary experimental hygiene rather than a new and expensive kind of animal work.

The distribution of the × marks is as informative as the totals. Calibration is the only item every class fails: not one of the audited canine systems reports probabilities that match observed frequencies, which is precisely the quantity a welfare or pain monitor has to threshold before it

Table 13: The ten reporting items applied to the four system classes audited in this survey. ✓ documented in the audited source; ∼ documented for some entries in the class; × absent or not reported; n/a the item does not bear on that class's claim. The final row counts ✓ over the items that apply.

Reporting item	**Consumer sentence apps**	**Rung-2 classifiers**	**Symbol-array studies**	**Open bioacoustic suites**
Split per dog and session declared	×	∼	n/a	✓
Error signal independent of the annotator	×	∼	✓	✓
Arousal baseline reported	×	×	n/a	n/a
Human raters blind to the label	×	∼	✓	n/a
Handler absent or blinded	×	×	✓	n/a
Calibration reported	×	×	×	×
Abstention reported	×	×	n/a	×
Negative controls included	×	×	✓	∼
Taxonomy declared before training	∼	∼	✓	✓
Claim written at the rung the evidence reaches	×	✓	✓	✓
Documented / applicable	**0/10**	**1/10**	**6/7**	**5/8**

can act. Conversely the two items the symbol-array studies clear most comfortably — handler blinding and a physically constrained alternative set — are the two that consumer products invert by design, since an app sold to a devoted owner in an uncontrolled home recruits the owner's expectation rather than removing it.

9.3 What adoption would buy

A shared benchmark changes the incentive structure, which is what the evaluation crisis actually is. Today the cheapest way to look good is to move the label, since labels are cheap and error signals are not; under this protocol the only way up the ladder is to acquire a better error signal, which means better experiments with dogs. The first version does not require new animal work: the existing falsifiable datasets — bark-context recordings, growl corpora, the button-array logs, the object-choice archives — already cover rungs 1–3, and a community release could stand them up in months. The hard part is institutional rather than technical, and it is the recommendation of this survey: a small consortium of the psychophysics groups that produced those corpora, a cross-species decoding lab, and one independent auditor should own the split, the taxonomies and the leaderboard, and vendor systems should be invited to submit to it rather than report numbers about themselves. Whether any of that happens is the real test of whether the field wants to know what it has built.

10 Distance, Agenda, and the Leap Question

We can now answer the title question without evasion. The answer is not a number of years but a map: the distance depends on which rung one means, and the rungs differ in kind (Table 14, Figure 5). Signal detection and affect classification are solved in the laboratory and shipping in products; the remaining work is engineering and validation, not discovery. A calibrated, validated welfare monitor reporting arousal and valence with honest uncertainty is plausibly a **2–4 year** horizon, with real payoff in early stress or pain detection, shelter welfare assessment, and veterinary triage. It is not "translation," but it is real. Intent inference on natural signals is

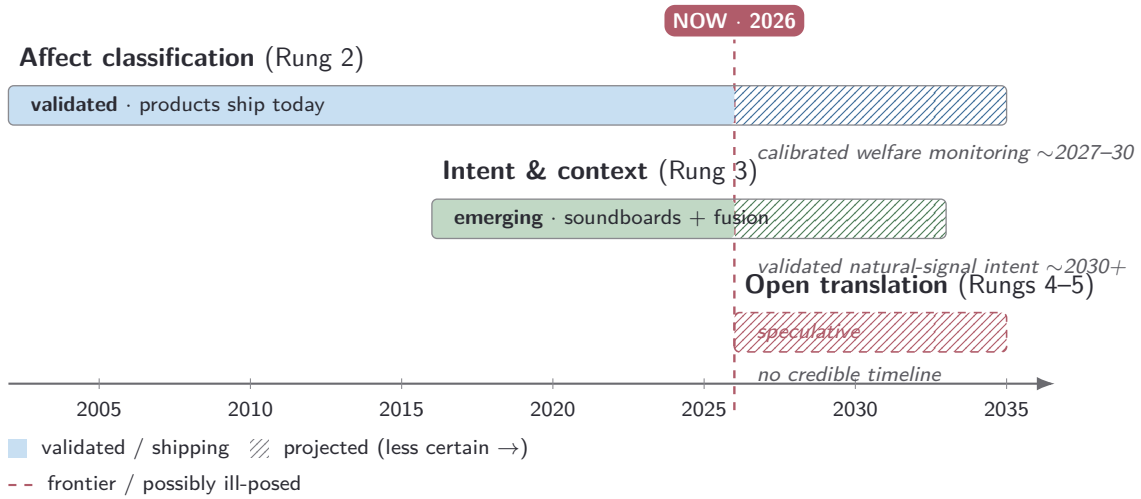


Figure 5: Capability roadmap, 2002–2035. Solid bars denote validated, shipping capability; hatched bars denote projection with confidence decreasing to the right. Track A (affect, Rung 2) heads toward calibrated welfare monitoring; Track B (intent, Rung 3) is emerging through soundboards and fusion, with validated natural-signal intent plausible after 2030; Track C (open translation, Rungs 4–5) has no credible timeline and may be permanently ill-posed. The dashed line marks the present (2026).

a research program, not a product cycle: the soundboard literature proves referential intent exists and is checkable, but with human-supplied symbols [15, 16]. Extending this to spontaneous barks, faces, and postures, we estimate **validated natural-signal intent at roughly 2030 or later**, contingent on the evaluation infrastructure being built. Open referential translation and free dialogue presuppose a generative symbolic system the biological evidence does not support; the honest answer is **no credible timeline**, because there may be no language there to translate. An engineered dialogue, structured human-to-dog input with the dog responding through trained symbols, is a Rung-3 artifact wearing Rung-5 clothing.

It is tempting to read the cross-species successes as evidence that scaling foundation models will eventually crack the dog. The reading is mistaken. CETI, DolphinGemma, and the finch models advance structure discovery and generation, the front of the pipeline; they do not supply semantics, and their authors say so [19, 21]. The canine bottleneck is at the back (grounding, validation) and, above Rung 3, in the subject itself. Scaling the encoder makes Rungs 1–2 better; it does not create a grammar that is not there. The productive reframe is to stop asking "when will we translate dogs?" and start asking "how well can we interpret canine state, with calibrated uncertainty and animal-grounded validation?" That question has answers, metrics, and welfare value, and it treats the dog as a signaler with its own modalities rather than a defective human whose sentences we are waiting to recover.

If that is right, the highest-leverage work is infrastructure and protocols, ordered as follows. First, a public, grounded canine corpus: synchronized audio, video, context metadata, and consequent behavior, with the soundboard citizen-science network as the natural collection substrate [15, 27]. Second, calibrated multimodal affect models fusing bark, growl, face, and posture on that corpus, with self-supervised encoders as the backbone [24, 25, 26]. Third, closed-loop intent validation: infer, act, and test the dog's response under blind conditions, the soundboard and CHAT protocols adapted to the home [16, 21]. Fourth, structure discovery with humility: apply the CETI and DolphinGemma toolkits to bark corpora with the explicit prior that graded displays may yield no clean alphabet, report structure as form rather than meaning, and heed the acoustic-similarity critique [30]; a negative result would itself sharpen the semantic-gap claim. Fifth, engineered rather than extracted dialogue: expand the soundboard into a richer negotiated symbol system, with the dog as partner and the behavior as check.

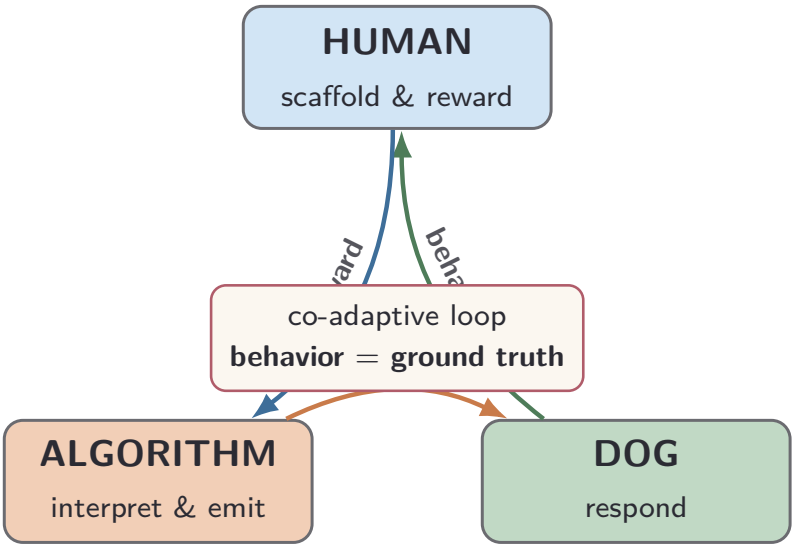


Figure 6: The developmental three-way closed loop. The algorithm emits a candidate signal or prompt; the dog responds with behavior (approach, look, press, refuse); the human supplies reward and blind validation. Each turn refreshes an animal-generated error signal, so a negotiated, grounded code can be built up over development—beginning in the puppy socialization window—rather than extracted from spontaneous vocalization.

10.1 The developmental closed loop: learning to talk together

The soundboard is an existence proof, and it points at a route the ladder alone does not name. A dog pressed a button and a human read the intent; the loop closed because a person supplied the meaning and the reward [15, 16]. What if the loop were tightened, automated, and begun early—algorithm, dog, and human each adapting to the others, a puppy learning a negotiated symbol system the way a child learns its first words? This is not decoding the bark. It is building, in real time, a shared code the bark never contained (Figure 6).

Three literatures converge on the design. The oldest is learning theory: behavior is shaped by its consequences, and a timely, consistent, unambiguous reinforcer—the clicker is the canonical human–dog implementation—turns spontaneous action into intentional communication [227, 228]. The second is development. Dogs pass through a primary socialization window in which human social signals are absorbed with unusual ease [229, 230, 231], and they are born ready for it: eight-week-old puppies follow human pointing and gaze with little training, a sensitivity that is largely heritable [232, 233] and that domestication, not learning alone, appears to have tuned [191, 192, 253]. The third is machine learning that treats feedback as a first-class signal: shaping an agent with a teacher's evaluative reward—TAMER, and the preference-based reinforcement underlying modern alignment—is exactly a closed loop in which reward stands in for a missing objective [235, 236]. An animal–computer interface delivering the clicker's consistency at machine rate and calibration is the obvious canine transposition [234].

Formally, the loop is a signaling game with a real animal as one player [237]: symbols are not mined from the dog's vocabulary but negotiated until they are grounded in shared consequences [238, 185, 186], and multi-agent learning shows a communicative code emerging between partners who must cooperate to be rewarded [239, 240]. The precedent is the great-ape rearing programs, begun in infancy and sustained by a closed loop of model, cue, and reward [241, 242];

their sober ending—symbol use without generative syntax [243]—is the caution that travels with the promise, and it is the same ceiling the ladder predicts.

Why the route matters to this survey's argument is one word: *error signal*. The field's central deficit is that a translation has no ground truth to be scored against (Section 8). A developmental closed loop does not find that ground truth in the bark; it manufactures one. The dog's consequent behavior is a genuine, animal-generated check on the system's hypothesis, refreshed every trial, and the human's reward calibrates it. This is the strongest available version of engineered, not extracted, dialogue, and the most concrete candidate for the field's attention-moment: not a new architecture, but a protocol that finally lets the dog correct us.

The limits travel with it. What is built is a pidgin—a small, human-scaffolded, referential code—not the transcription of spontaneous vocalization; the semantic gap is bypassed by construction, not crossed [187, 189]. Rearing a puppy into an instrumented training regime is a welfare question before it is a technical one, and the animal's consent is the constraint the ethics framework refuses to trade away [234, 201]. A dog raised inside a well-designed closed loop could become the most legible partner we have ever built—fluent in a language two species invented for one. That is worth doing, and it is, precisely, not the fantasy in the app store.

10.2 Could there be an "Attention Is All You Need" moment?

The Transformer arrived as a machine-translation paper and became a general-purpose engine of the field [44], so it is natural to ask whether one architectural insight could produce the same leap for canine decoding. The history of 2017 is usually told as a story about architecture; it is equally a story about preconditions. Attention caught fire because three conditions were already in place: human languages supply a productive, recursive target; web-scale parallel corpora (Europarl, OPUS) supplied supervision at scale [53, 54]; and reference translations gave the field a cheap shared error signal, BLEU, against which thousands of groups could iterate [55]. The architecture was the spark; target structure, parallel data, and a trustworthy error signal were the dry tinder.

Auditing the canine case against those conditions is sobering. The biological record shows graded, context-bound displays and no evidenced generative grammar, so the first condition is precisely what fails [1, 4]. A bark–sentence bitext cannot be harvested from the web; it can only be constructed, and the soundboard is the first small constructed one [15]. The error signal is absent altogether, which is why the field cannot compound its progress. A new architecture, however elegant, lands on wet tinder.

Yet the same history says that leaps begin when a representation makes hidden structure visible: the CETI alphabet multiplied the inventory of distinguishable coda forms nearly tenfold [19], and a comparable discovery of discrete, context-grounded units in canine displays would rewrite our ladder. The more probable attention-moment is infrastructural: a grounded public corpus plus blind, closed-loop validation would supply the error signal that BLEU supplied to MT, with the dog's own behavior as the reference translation [16, 21]. When a field acquires a trustworthy error signal, progress begins to compound. So yes: a leap is possible, and the door should stay open to empirical surprise. It will arrive as a discovery about the dog, or as a protocol that lets the dog correct us, recognizable by the same signature as 2017: suddenly, everyone can measure progress. Until then, the honest posture is the one any good listener adopts toward a companion whose language is not our own: attention, patience, and the willingness to be corrected.

10.3 What would falsify our thesis

The central claim, that Rungs 4–5 are ill-posed for natural canine communication, would be undermined by a reproducible demonstration of a productive, open referential vocabulary in spontaneous dog signals validated by playback, or by evidence of recursive, generative combination in natural

canine displays (not trained soundboard sequences). Absent such evidence, the ladder stands. We invite the community to try to falsify it; that is how the distance estimate should shrink.

Table 14: Distance estimate by rung. Horizons are judgments from the evidence reviewed, not predictions.

Rung	Status	Horizon	Dominant blocker
1 Signal	solved	now	engineering only
2 Affect	shipping	2–4 yr to validated welfare use	calibration, public benchmarks
3 Intent	foothold	∼2030+	grounded, validated data
4 Reference	unproven	no credible timeline	no productive symbol system
5 Dialogue	possibly ill-posed	possibly never	no evidenced generative grammar

11 Ethics: The Animal as Partner

A technology that claims to read an animal's mind carries obligations an ordinary classifier does not. The most immediate is welfare: an owner told "she says she loves you" may overlook the flattened ears, the whale eye, the lip lick that signal stress, and an owner told "my tummy hurts" may over-react or dismiss genuine pain when the app stays silent. Human observers already project their categories onto dogs, reading a stress-driven appeasement face as guilt [9, 254], and a resonant language model amplifies the projection. Welfare-first design therefore surfaces calibrated state estimates with uncertainty rather than sentences, never overrides observable body signals, routes high-stakes inferences (pain, aggression, distress) to human or veterinary review, and collects passively, without induced stress to elicit training barks.

Honesty follows from the evaluation crisis: separate the calibrated classifier from the narrative interface, label the latter as interpretation, disclose that accuracy figures refer to emotion classification, and refrain from unvalidated reverse-direction claims. Privacy follows from the always-on home sensor: on-device inference where possible (the DolphinGemma-on-Pixel precedent is instructive [21]), minimal retention, explicit household consent for cloud training, and clear deletion. The animal cannot consent; the household must, and the design should make that consent meaningful.

Finally, a conceptual obligation. Treating the dog as an oracle whose "words" we extract flattens its communicative life into a human format. The soundboard paradigm is ethically attractive precisely because it treats the dog as a partner in a negotiated symbol system, with the dog's behavior as the check [15, 16]; CETI's leadership has likewise stressed that breakthroughs in whale communication should be a collective human decision, with responsible-interaction guidelines before signals are sent to wild animals. This framework inherits the consensus principles of responsible AI (transparency, justice, non-maleficence, responsibility, privacy) [196], the documented gap between principles and practice [197, 198], the political-economy critique of hidden data and labor supply chains [199], the risk-tiered regulatory baseline of the EU AI Act [200], and the concrete engineering of fair, safe, and human-compatible systems—disparate-impact auditing, specification-gaming avoidance, and compatibility [255, 257, 256]. On the animal side, the precautionary principle for sentience warrants moral consideration under uncertainty [202], and recent work locates the edge of sentience in exactly the graded affective capacities this survey treats as decodable [201]. The synthesis is a double obligation: be honest about what the model knows, and be careful toward what the animal feels.

For builders and reviewers, the framework condenses to six checks: report Rung-2 metrics on public held-out data with calibration, never as "translation accuracy"; separate and label interpretation versus measurement; validate intent claims closed-loop and blind or call them hypotheses;

collect passively, infer on-device, minimize retention, obtain meaningful consent; route high-stakes outputs to human or veterinary review and never override body signals; and treat the animal as a partner, keeping welfare as the terminal criterion. None of this slows the useful science; all of it protects the animals and the credibility of the field.

12 Conclusion

We began with a folkloric question, how far are we from translating what a dog says, and end with a more useful answer. "Translation" is not one capability but five, and the distance to each is different in kind. Signal detection and affect classification (Rungs 1–2) are here and shipping; validated affect-and-intent inference (Rung 3) is a research program with a foothold in the soundboard literature and a plausible horizon around 2030; open referential translation and free dialogue (Rungs 4–5) presuppose a generative symbolic system the biological evidence does not support, and may be permanently ill-posed. The canine Rosetta Stone is a metaphor with a catch: the original worked because one decree existed in three scripts, and for the dog there may be no decree to transcribe, only a rich, graded, affective life we can learn to read but not to render as sentences.

Anyone who has lived with a dog knows the scene that motivates this literature: the pause at the door, the whine at the empty bowl, the look that seems to wait for an answer. The information in that moment is real, affect, arousal, intent, addressed to us. What the evidence does not support is the next step of the fantasy, a sentence inside the whine waiting for a decoder. Honoring the moment means reading it well, with calibrated uncertainty, and letting the dog's subsequent behavior correct us. That is a research program, and it is also a form of respect.

Three findings anchor the verdict. The biology: dogs signal affect, arousal, identity, motivation, and, with human scaffolding, referential intent, but they do not possess an open-ended generative grammar [1, 4, 6, 16]. The technology: the pipeline's front (encoding, structure discovery) is maturing fast on the cross-species foundation-model boom, while its back (grounding, validation) is where canine-specific work is needed and commercial effort is absent [24, 25, 19, 21]. The evaluation: headline accuracies are unfalsifiable as translation metrics because no ground-truth intended message exists to score against; they are, at best, Rung-2 emotion-classification numbers [5]. The market demonstrates Rung 2 and sells Rung 5, and the gap between those is the subject of this survey.

The way forward is better infrastructure and protocols, and Section 9 specifies one in enough detail to run: a public grounded canine corpus, calibrated multimodal affect models, closed-loop blind validation, and an ethics framework that keeps welfare as the terminal criterion. The achievable goal is interpretation with calibrated uncertainty, and it is within reach this decade. The dog at the door does not need a transcript. It needs a listener. A field that builds the instruments for that listening will have made the oldest conversation between our species a little more honest, and a little more kind.

12.1 Limitations of this survey

This is a narrative survey, not a PRISMA systematic review; sources were gathered through structured search in November 2026, and coverage is broad rather than exhaustive. Several fast-moving items (WhAM, ZF-AIM, vendor metrics) rest on project reports and journalism rather than peer review and are labeled as such; commercial accuracies are reported as claims and were not independently verified. Some citations were confirmed to venue, year, and DOI via secondary indexes and should be checked against the primary record before reuse. The distance estimates are expert judgments, not forecasts, and Section 10 states what would falsify them. The Canine Ladder Benchmark of Section 9 is a proposal specified from the published record rather than a leaderboard that has been run; its baselines become numbers only once someone executes them, and the rung-5

rejection test in particular is an argument about what could count as evidence, not a result. These limitations bound the central claim, which is the point.

Acknowledgments and tools

This survey was prepared with the assistance of open research-agent tooling, including the Scientific Agent Skills literature-review and autoresearch workflows [38], which contributed the search-documentation and citation-verification discipline used here. The figures are original TikZ constructions released with the repository.

Data and code availability

The LaTeX source, figures, bibliography, and project page are openly available in the `woof-to-word` repository, together with the search log, the script that regenerates the corpus statistics of Figure 1, and its per-entry CSV output: `https://github.com/mcxiaoxiao/woof2word`. Corrections and falsification attempts are welcome.

References

[1] P. Pongrácz, C. Molnár, A. Miklósi, and V. Csányi, "Human listeners are able to classify dog (*Canis familiaris*) barks recorded in different situations," *Journal of Comparative Psychology*, vol. 119, no. 2, pp. 136–144, 2005.

[2] P. Pongrácz, C. Molnár, and A. Miklósi, "Acoustic parameters of dog barks carry emotional information for humans," *Applied Animal Behaviour Science*, vol. 100, no. 3–4, pp. 228–240, 2006.

[3] C. Molnár, P. Pongrácz, A. Dóka, and A. Miklósi, "Can humans discriminate between dogs on the basis of the acoustic parameters of barks?" *Behavioural Processes*, vol. 73, no. 1, pp. 76–83, 2006.

[4] T. Faragó, N. Takács, A. Miklósi, and P. Pongrácz, "Dog growls express various contextual and affective content for human listeners," *Royal Society Open Science*, vol. 4, p. 170134, 2017.

[5] "What is my dog trying to tell me? The automatic recognition of the context and perceived emotion of dog barks," in *Proc. IEEE ICASSP*, 2018. DOI: 10.1109/ICASSP.2018.8461757.

[6] B. M. Waller, K. Peirce, C. C. Caeiro, et al., "DogFACS: The dog facial action coding system," 2013. Available: `https://www.animalfacs.com`.

[7] J. Kaminski, J. Waller, R. Diogo, A. Hartstone-Rose, and A. M. Burrows, "Human attention affects facial expressions in domestic dogs," *Scientific Reports*, vol. 7, p. 13066, 2017.

[8] "Domestication constrains the ability of dogs to convey emotions via facial expressions in comparison to their wolf ancestors," *Scientific Reports*, 2024. DOI: 10.1038/s41598-024-61110-6.

[9] "Explainable automated recognition of emotional states from canine facial expressions," *Scientific Reports*, 2022. DOI: 10.1038/s41598-022-27079-w.

[10] "Pet dog facial expression recognition based on convolutional neural network and improved whale optimization algorithm," *Scientific Reports*, 2023. DOI: 10.1038/s41598-023-30442-0.

[11] "Automated analysis of emotional expressions in dogs," *Scientific Reports*, 2025. DOI: 10.1038/s41598-025-15741-y.

[12] "Dog facial landmarks detection and its applications for facial analysis," *Scientific Reports*, 2025. DOI: 10.1038/s41598-025-07040-3.

[13] J. Kaminski, J. Call, and J. Fischer, "Word learning in a domestic dog: Evidence for "fast mapping"," *Science*, vol. 304, no. 5677, pp. 1682–1683, 2004.

[14] J. W. Pilley and A. K. Reid, "Border collie comprehends object names as verbal referents," *Behavioural Processes*, vol. 86, no. 2, pp. 184–195, 2011.

[15] A. P. M. Bastos, A. Evenson, P. M. Wood, et al., "How do soundboard-trained dogs respond to human button presses? An investigation into word comprehension," *PLoS ONE*, vol. 19, p. e0307189, 2024. DOI: 10.1371/journal.pone.0307189.

[16] A. P. M. Bastos, Z. N. Houghton, L. Naranjo, and F. Rossano, "Soundboard-trained dogs produce non-accidental, non-random and non-imitative two-button combinations," *Scientific Reports*, vol. 14, p. 28771, 2024. DOI: 10.1038/s41598-024-79517-6.

[17] "Let's talk about "talking" dogs! Reviewing the science of soundboard use," *Animal Cognition*, 2025. PMC12367964.

[18] A. P. M. Bastos and F. Rossano, "Soundboard-using pets?" *Interaction Studies*, vol. 24, no. 2, pp. 311–334, 2023. DOI: 10.1075/is.22050.pin.

[19] P. Sharma, S. Gero, R. Payne, D. F. Gruber, D. Rus, A. Torralba, and J. Andreas, "Contextual and combinatorial structure in sperm whale vocalisations," *Nature Communications*, vol. 15, p. 3617, 2024. DOI: 10.1038/s41467-024-47221-8.

[20] J. Andreas, G. Beguš, M. M. Bronstein, et al., "Toward understanding the communication in sperm whales," *iScience*, vol. 25, p. 104393, 2022.

[21] Google DeepMind, Georgia Tech, and Wild Dolphin Project, "DolphinGemma: How Google AI is helping decode dolphin communication," Project report, 14 April 2025.

[22] Project CETI, "WhAM: Whale Acoustics Model," Technical report, 2025.

[23] "Towards a translative model of sperm whale vocalization," *arXiv:2512.02206*, 2025.

[24] M. Hagiwara, "AVES: Animal vocalization encoder based on self-supervision," in *Proc. IEEE ICASSP*, 2023.

[25] D. Robinson, M. Miron, M. Hagiwara, and O. Pietquin, "NatureLM-audio: An audio–language foundation model for bioacoustics," *arXiv:2411.07186*, 2024.

[26] B. Ghani, T. Denton, S. Kahl, and H. Klinck, "Global birdsong embeddings enable superior transfer learning for bioacoustic classification," *Scientific Reports*, vol. 13, p. 22876, 2023.

[27] M. Hagiwara et al., "BEANS: The benchmark of animal sounds," in *Proc. IEEE ICASSP*, 2023.

[28] "OpenBEATs: A fully open-source general-purpose audio encoder," *arXiv:2507.14129*, 2025.

[29] Earth Species Project and McGill University, "ZF-AIM: An interactive zebra-finch vocalization model," Technical report, 2026 (not yet peer reviewed).

[30] Y. Yovel, Commentary on acoustic-similarity pitfalls in animal-language models, reported in *Nature* news coverage, 2025–2026.

[31] Traini (PetGPT / PEBI model), Commercial pet-empathy AI, 2024–2026. Vendor-reported metrics; see Section 8.

[32] Takara Co., "Bow-Lingual" dog-bark translator, 2002. Ig Nobel Peace Prize, 2002.

[33] J. Sanchez (Akvelon), "MeowTalk" cat-vocalization app, 2020.

[34] "Meng Xiao Yi" pet translator (Alibaba Qwen-based; vendor-claimed 94.6%), 2026. Reported via Chinese technology press.

[35] Petpuls AI collar (Seoul National University spin-off), 2021.

[36] A. Quaranta, M. Siniscalchi, and G. Vallortigara, "Asymmetric tail-wagging responses by dogs to different emotive stimuli," *Current Biology*, vol. 17, no. 6, pp. R199–R201, 2007.

[37] A. Andics, A. Gácsi, T. Faragó, A. Kis, and A. Miklósi, "Voice-sensitive regions in the dog and human brain are revealed by comparative fMRI," *Current Biology*, vol. 24, no. 5, pp. 574–578, 2014.

[38] T. Kassis, V. Agarwal, Y. He, D. Patel, and A. M. Brueckner, "Scientific Agent Skills: A library of procedural knowledge for research agents," *arXiv:2609.00065*, 2026. DOI: 10.48550/arXiv.2609.00065.

[39] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to sequence learning with neural networks," in *NeurIPS*, 2014.

[40] K. Cho et al., "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *EMNLP*, 2014.

[41] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," in *ICLR*, 2015.

[42] M.-T. Luong, H. Pham, and C. D. Manning, "Effective approaches to attention-based neural machine translation," in *EMNLP*, 2015.

[43] J. Gehring et al., "Convolutional sequence to sequence learning," in *ICML*, 2017.

[44] A. Vaswani et al., "Attention is all you need," in *NeurIPS*, 2017. arXiv:1706.03762.

[45] Y. Wu et al., "Google's neural machine translation system: Bridging the gap between human and machine translation," arXiv:1609.08144, 2016.

[46] M. Johnson et al., "Google's multilingual neural machine translation system: Enabling zero-shot translation," *TACL*, vol. 5, pp. 339–351, 2017.

[47] O. Firat, K. Cho, and Y. Bengio, "Multi-way, multilingual neural machine translation with a shared attention mechanism," in *NAACL*, 2016.

[48] M. Ott et al., "Scaling neural machine translation," in *WMT*, 2018.

[49] S. Edunov et al., "Understanding back-translation at scale," in *EMNLP*, 2018.

[50] R. Sennrich, B. Haddow, and A. Birch, "Neural machine translation of rare words with subword units," in *ACL*, 2016.

[51] R. Sennrich, B. Haddow, and A. Birch, "Improving neural machine translation models with monolingual data," in *ACL*, 2016.

[52] P. Koehn et al., "Moses: Open source toolkit for statistical machine translation," in *ACL (demo)*, 2005.

[53] P. Koehn, "Europarl: A parallel corpus for statistical machine translation," in *MT Summit*, 2005.

[54] J. Tiedemann, "Parallel data, tools and interfaces in OPUS," in *LREC*, 2012.

[55] K. Papineni et al., "BLEU: A method for automatic evaluation of machine translation," in *ACL*, 2002.

[56] M. Denkowski and A. Lavie, "Meteor universal: Language specific translation evaluation for any target language," in *EACL*, 2014.

[57] R. Rei, C. Stewart, and A. Farinha, "Unbabel's participation in the WMT20 metrics shared task (COMET)," in *WMT*, 2020.

[58] M. Freitag et al., "Experts, errors, and context: A large-scale study of human evaluation for machine translation," *TACL*, vol. 9, pp. 146–164, 2021.

[59] O. Bojar et al., "Findings of the 2018 conference on machine translation (WMT18)," in *WMT*, 2018.

[60] L. Barrault et al., "Findings of the 2019 conference on machine translation (WMT19)," in *WMT*, 2019.

[61] T. Kocmi et al., "Findings of the 2022 conference on machine translation (WMT22)," in *WMT*, 2022.

[62] M. R. Costa-jussà et al., "No language left behind: Scaling human-centered machine translation," arXiv:2207.04672, 2022.

[63] NLLB Team, "Scaling neural machine translation to 200 languages," *Nature*, vol. 630, pp. 841–846, 2024.

[64] A. Fan et al., "Beyond English-centric multilingual machine translation," *JMLR*, vol. 22, pp. 1–48, 2021.

[65] M. Artetxe et al., "Unsupervised neural machine translation," in *ICLR*, 2018.

[66] G. Lample et al., "Phrase-based & neural unsupervised machine translation," in *EMNLP*, 2018.

[67] H. Schwenk et al., "WikiMatrix: Mining 135M parallel sentences in 1620 language pairs from Wikipedia," in *ACL*, 2021.

[68] A. El-Kishky et al., "CCAligned: A massive collection of cross-lingual web-document pairs," in *LREC*, 2020.

[69] A. Bérard et al., "Listen and translate: A proof of concept for end-to-end speech-to-text translation," in *NeurIPS Workshop*, 2016.

[70] R. J. Weiss et al., "Sequence-to-sequence models can directly translate foreign speech," in *Interspeech*, 2017.

[71] Y. Jia et al., "Direct speech-to-speech translation with a sequence-to-sequence model," in *Interspeech*, 2019.

[72] Seamless Communication et al., "SeamlessM4T: Massively multilingual & multimodal machine translation," arXiv:2308.11596, 2023.

[73] Seamless Communication et al., "Seamless: Multilingual expressive and streaming speech translation," arXiv:2312.05187, 2023.

[74] Y. Liu et al., "Multilingual denoising pre-training for neural machine translation (mBART)," *TACL*, vol. 8, pp. 726–742, 2020.

[75] T. Mikolov et al., "Efficient estimation of word representations in vector space," arXiv:1301.3781, 2013.

[76] J. Pennington, R. Socher, and C. D. Manning, "GloVe: Global vectors for word representation," in *EMNLP*, 2014.

[77] M. E. Peters et al., "Deep contextualized word representations (ELMo)," in *NAACL*, 2018.

[78] J. Howard and S. Ruder, "Universal language model fine-tuning for text classification (ULMFiT)," in *ACL*, 2018.

[79] J. Devlin et al., "BERT: Pre-training of deep bidirectional transformers for language understanding," in *NAACL*, 2019. arXiv:1810.04805.

[80] A. Radford et al., "Improving language understanding by generative pre-training," OpenAI, Tech. Rep., 2018.

[81] A. Radford et al., "Language models are unsupervised multitask learners," OpenAI, Tech. Rep., 2019.

[82] T. Brown et al., "Language models are few-shot learners," in *NeurIPS*, 2020. arXiv:2005.14165.

[83] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer (T5)," *JMLR*, vol. 21, pp. 1–67, 2020.

[84] M. Lewis et al., "BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension," in *ACL*, 2020.

[85] Y. Liu et al., "RoBERTa: A robustly optimized BERT pretraining approach," arXiv:1907.11692, 2019.

[86] Z. Yang et al., "XLNet: Generalized autoregressive pretraining for language understanding," in *NeurIPS*, 2019.

[87] K. Clark et al., "ELECTRA: Pre-training text encoders as discriminators rather than generators," in *ICLR*, 2020.

[88] P. He et al., "DeBERTa: Decoding-enhanced BERT with disentangled attention," in *ICLR*, 2021.

[89] A. Conneau et al., "Unsupervised cross-lingual representation learning at scale (XLM-R)," in *ACL*, 2020.

[90] T. Pires, E. Schlinger, and D. Garrette, "How multilingual is multilingual BERT?" in *ACL*, 2019.

[91] A. Conneau and G. Lample, "Cross-lingual language model pretraining (XLM)," in *NeurIPS*, 2019.

[92] H. Touvron et al., "LLaMA: Open and efficient foundation language models," arXiv:2302.13971, 2023.

[93] H. Touvron et al., "Llama 2: Open foundation and fine-tuned chat models," arXiv:2307.09288, 2023.

[94] OpenAI, "GPT-4 technical report," arXiv:2303.08774, 2023.

[95] Gemini Team, "Gemini: A family of highly capable multimodal models," arXiv:2312.11805, 2024.

[96] Anthropic, "The Claude 3 model family: Opus, Sonnet, Haiku," Tech. Rep., 2024.

[97] Gemma Team, "Gemma: Open models based on Gemini research and technology," arXiv:2403.08295, 2024.

[98] L. Ouyang et al., "Training language models to follow instructions with human feedback (InstructGPT)," in *NeurIPS*, 2022.

[99] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," in *NeurIPS*, 2022.

[100] R. Bommasani et al., "On the opportunities and risks of foundation models," arXiv:2108.07258, 2021.

[101] J. Kaplan et al., "Scaling laws for neural language models," arXiv:2001.08361, 2020.

[102] J. Hoffmann et al., "Training compute-optimal large language models (Chinchilla)," in *NeurIPS*, 2022.

[103] A. Radford et al., "Learning transferable visual models from natural language supervision (CLIP)," in *ICML*, 2021.

[104] J.-B. Alayrac et al., "Flamingo: A visual language model for few-shot learning," in *NeurIPS*, 2022.

[105] H. Liu et al., "Visual instruction tuning (LLaVA)," in *NeurIPS*, 2023.

[106] J. Li et al., "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *ICML*, 2023.

[107] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," arXiv:1807.03748, 2018.

[108] S. Schneider et al., "wav2vec: Unsupervised pre-training for speech recognition," in *Interspeech*, 2019.

[109] A. Baevski et al., "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *NeurIPS*, 2020. arXiv:2006.11477.

[110] W.-N. Hsu et al., "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM TASLP*, vol. 29, pp. 3451–3460, 2021.

[111] S. Chen et al., "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE JSTSP*, vol. 16, no. 6, pp. 1505–1518, 2022.

[112] A. Baevski et al., "data2vec: A general framework for self-supervised learning in speech, vision and language," in *ICML*, 2022.

[113] A. Baevski et al., "Efficient self-supervised learning with contextualized target representations (data2vec 2.0)," in *ICML*, 2023.

[114] A. Radford et al., "Robust speech recognition via large-scale weak supervision (Whisper)," in *ICML*, 2023. arXiv:2212.04356.

[115] A. Graves et al., "Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks," in *ICML*, 2006.

[116] A. Graves, A. Mohamed, and G. Hinton, "Speech recognition with deep recurrent neural networks," in *ICASSP*, 2013.

[117] D. Amodei et al., "Deep Speech 2: End-to-end speech recognition in English and Mandarin," in *ICML*, 2016.

[118] W. Chan et al., "Listen, attend and spell: A neural network for large vocabulary conversational speech recognition," in *ICASSP*, 2016.

[119] A. Gulati et al., "Conformer: Convolution-augmented transformer for speech recognition," in *Interspeech*, 2020.

[120] V. Panayotov et al., "Librispeech: An ASR corpus based on public domain audio books," in *ICASSP*, 2015.

[121] R. Ardila et al., "Common Voice: A massively-multilingual speech corpus," in *LREC*, 2020.

[122] V. Pratap et al., "MLS: A large-scale multilingual dataset for speech research," in *Interspeech*, 2020.

[123] J. S. Garofolo et al., "TIMIT acoustic-phonetic continuous speech corpus," Linguistic Data Consortium, 1993.

[124] J. J. Godfrey, E. C. Holliman, and J. McDaniel, "SWITCHBOARD: Telephone speech corpus for research and development," in *ICASSP*, 1992.

[125] S.-w. Yang et al., "SUPERB: Speech processing universal performance benchmark," in *Interspeech*, 2021.

[126] A. Baevski et al., "Unsupervised speech recognition," in *NeurIPS*, 2021.

[127] A. Babu et al., "XLS-R: Self-supervised cross-lingual speech representation learning at scale," in *Interspeech*, 2022.

[128] V. Pratap et al., "Scaling speech technology to 1,000+ languages (MMS)," in *ACL*, 2023.

[129] Y. Zhang et al., "Google USM: Scaling automatic speech recognition beyond 100 languages," arXiv:2303.01037, 2023.

[130] C. Wang et al., "CoVoST: A diverse multilingual speech-to-text translation corpus," in *LREC*, 2020.

[131] C. Wang et al., "CoVoST 2 and massively multilingual speech-to-text translation," in *Interspeech*, 2021.

[132] A. van den Oord et al., "WaveNet: A generative model for raw audio," arXiv:1609.03499, 2016.

[133] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, "Neural discrete representation learning (VQ-VAE)," in *NeurIPS*, 2017.

[134] P. Dhariwal et al., "Jukebox: A generative model for music," arXiv:2005.00341, 2020.

[135] N. Zeghidour et al., "SoundStream: An end-to-end neural audio codec," *IEEE/ACM TASLP*, vol. 30, pp. 495–507, 2021.

[136] A. Défossez et al., "High fidelity neural audio compression (EnCodec)," *TMLR*, 2023. arXiv:2210.13438.

[137] Z. Borsos et al., "AudioLM: A language modeling approach to audio generation," *IEEE/ACM TASLP*, vol. 31, pp. 252–264, 2023.

[138] A. Agostinelli et al., "MusicLM: Generating music from text," arXiv:2301.11325, 2023.

[139] J. Copet et al., "Simple and controllable music generation (MusicGen)," in *NeurIPS*, 2023.

[140] F. Kreuk et al., "AudioGen: Textually guided audio generation," in *ICLR*, 2023.

[141] C. Wang et al., "Neural codec language models are zero-shot text to speech synthesizers (VALL-E)," arXiv:2301.02111, 2023.

[142] D. Yang et al., "UniAudio: An audio foundation model toward universal audio generation," in *ICML*, 2024.

[143] P. K. Rubenstein et al., "AudioPaLM: A large language model that can speak and listen," arXiv:2306.12925, 2023.

[144] C. Tang et al., "SALMONN: Towards generic hearing abilities for large language models," in *ICLR*, 2024.

[145] Y. Chu et al., "Qwen-Audio: Advancing universal audio understanding via unified large-scale audio-language models," arXiv:2311.07919, 2023.

[146] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE TKDE*, vol. 22, no. 10, pp. 1345–1359, 2010.

[147] F. Zhuang et al., "A comprehensive survey on transfer learning," *Proc. IEEE*, vol. 109, no. 1, pp. 43–76, 2021.

[148] J. Yosinski et al., "How transferable are features in deep neural networks?" in *NeurIPS*, 2014.

[149] E. M. Bender and A. Koller, "Climbing towards NLU: On meaning, form, and understanding in the age of data," in *ACL*, 2020.

[150] E. M. Bender et al., "On the dangers of stochastic parrots: Can language models be too big?" in *FAccT*, 2021.

[151] T. Gebru et al., "Datasheets for datasets," *CACM*, vol. 64, no. 12, pp. 86–92, 2021.

[152] M. Mitchell et al., "Model cards for model reporting," in *FAT**, 2019.

[153] P. Liang et al., "Holistic evaluation of language models (HELM)," *TMLR*, 2023.

[154] A. Srivastava et al., "Beyond the imitation game: Quantifying and extrapolating the capabilities of language models (BIG-bench)," *TMLR*, 2023.

[155] D. Hendrycks et al., "Measuring massive multitask language understanding (MMLU)," in *ICLR*, 2021.

[156] S. Lin, J. Hilton, and O. Evans, "TruthfulQA: Measuring how models mimic human falsehoods," in *ACL*, 2022.

[157] Z. Ji et al., "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, pp. 1–38, 2023.

[158] J. Maynez et al., "On faithfulness and factuality in abstractive summarization," in *ACL*, 2020.

[159] S. Harnad, "The symbol grounding problem," *Physica D*, vol. 42, pp. 335–346, 1990.

[160] Y. Bisk et al., "Experience grounds language," in *EMNLP*, 2020.

[161] D. Hupkes et al., "Compositionality decomposed: How do neural networks generalise?" *JAIR*, vol. 67, pp. 757–795, 2020.

[162] B. M. Lake and M. Baroni, "Human-like systematic generalization through a meta-learning neural network," *Nature*, vol. 623, pp. 115–121, 2023.

[163] B. M. Lake and M. Baroni, "Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks," in *ICML*, 2018.

[164] A. Conneau et al., "What you can cram into a single vector: Probing sentence embeddings for linguistic properties," in *ACL*, 2018.

[165] A. Rogers, O. Kovaleva, and A. Rumshisky, "A primer on BERTology: What we know about how BERT works," *TACL*, vol. 8, pp. 842–866, 2020.

[166] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *ICML*, 2017. arXiv:1706.04599.

[167] D. Snyder, D. Garcia-Romero, G. Sell, D. Povey, and S. Khudanpur, "X-vectors: Robust DNN embeddings for speaker recognition," in *Proc. ICASSP*, 2018, pp. 5329–5333.

[168] L. Proops, K. McComb, and D. Reby, "Cross-modal individual recognition in domestic horses (*Equus caballus*)," *PNAS*, vol. 106, no. 4, pp. 947–951, 2009. DOI 10.1073/pnas.0809127105.

[169] L. Proops and K. McComb, "Cross-modal individual recognition in domestic horses extends to familiar humans," *Proc. R. Soc. B*, vol. 279, pp. 3131–3138, 2012. DOI 10.1098/rspb.2012.0626.

[170] S. Kahl et al., "BirdNET: A deep learning solution for avian diversity monitoring," *Ecological Informatics*, vol. 61, p. 101236, 2021.

[171] D. Stowell, "Computational bioacoustics with deep learning: A review and roadmap," *PeerJ*, vol. 10, p. e13152, 2022.

[172] D. Stowell and M. D. Plumbley, "Automatic large-scale classification of bird sounds is strongly improved by unsupervised feature learning," *PeerJ*, vol. 2, p. e488, 2014.

[173] V. M. Janik and P. J. B. Slater, "Context-specific use suggests that bottlenose dolphin signature whistles are cohesion calls," *Animal Behaviour*, vol. 56, pp. 829–838, 1998.

[174] S. L. King and V. M. Janik, "Bottlenose dolphins can use learned vocal labels to address each other," *PNAS*, vol. 110, pp. 13216–13221, 2013.

[175] R. M. Seyfarth, D. L. Cheney, and P. Marler, "Monkey responses to three different alarm calls: Evidence of predator classification and semantic communication," *Science*, vol. 210, pp. 801–803, 1980.

[176] D. L. Cheney and R. M. Seyfarth, *How Monkeys See the World*. University of Chicago Press, 1990.

[177] K. von Frisch, *The Dance Language and Orientation of Bees*. Harvard University Press, 1967.

[178] C. Crockford et al., "Wild chimpanzees inform ignorant group members of danger," *Current Biology*, vol. 22, pp. 142–146, 2012.

[179] T. N. Suzuki, D. Wheatcroft, and M. Griesser, "Experimental evidence for compositional syntax in bird calls," *Nature Communications*, vol. 7, p. 10986, 2016.

[180] P. Schlenker et al., "Formal monkey linguistics," *Theoretical Linguistics*, vol. 42, pp. 1–90, 2016.

[181] C. N. Slobodchikoff, A. Paseka, and J. L. Verdolin, "Prairie dog alarm calls encode labels about predator colors," *Animal Cognition*, vol. 12, pp. 435–439, 2009.

[182] C. S. Evans, L. Evans, and P. Marler, "On the meaning of alarm calls: Functional reference in an avian vocal system," *Animal Behaviour*, vol. 46, pp. 23–38, 1993.

[183] M. D. Hauser, *The Evolution of Communication*. MIT Press, 1996.

[184] M. Tomasello, *Origins of Human Communication*. MIT Press, 2008.

[185] S. Kirby, H. Cornish, and K. Smith, "Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language," *PNAS*, vol. 105, pp. 10681–10686, 2008.

[186] K. Smith, S. Kirby, and H. Brighton, "Iterated learning: A framework for the emergence of language," *Artificial Life*, vol. 9, pp. 371–386, 2003.

[187] M. D. Hauser, N. Chomsky, and W. T. Fitch, "The faculty of language: What is it, who has it, and how did it evolve?" *Science*, vol. 298, pp. 1569–1579, 2002.

[188] S. Pinker, *The Language Instinct*. William Morrow, 1994.

[189] N. Chomsky, *Syntactic Structures*. Mouton, 1957.

[190] G. K. Zipf, *Human Behavior and the Principle of Least Effort*. Addison-Wesley, 1949.

[191] B. Hare et al., "The domestication of social cognition in dogs," *Science*, vol. 298, pp. 1634–1636, 2002.

[192] A. Miklósi et al., "A simple reason for a big difference: Wolves do not look back at humans, but dogs do," *Current Biology*, vol. 13, pp. 763–766, 2003.

[193] K. Soproni et al., "Comprehension of human communicative signs in pet dogs," *Journal of Comparative Psychology*, vol. 115, pp. 122–126, 2001.

[194] G. S. Berns et al., "Functional MRI in awake unrestrained dogs," *PLoS ONE*, vol. 7, p. e38027, 2012.

[195] L. A. F. Frantz et al., "Genomic and archaeological evidence suggest a dual origin of domestic dogs," *Science*, vol. 352, pp. 1228–1231, 2016.

[196] A. Jobin, M. Ienca, and E. Vayena, "The global landscape of AI ethics guidelines," *Nature Machine Intelligence*, vol. 1, pp. 389–399, 2019.

[197] B. Mittelstadt, "Principles alone cannot guarantee ethical AI," *Nature Machine Intelligence*, vol. 1, pp. 501–507, 2019.

[198] L. Floridi et al., "AI4People—An ethical framework for a good AI society," *Minds and Machines*, vol. 28, pp. 689–707, 2018.

[199] K. Crawford, *Atlas of AI*. Yale University Press, 2021.

[200] European Parliament, "Artificial Intelligence Act," Regulation (EU) 2024/1689, 2024.

[201] J. Birch, *The Edge of Sentience*. Oxford University Press, 2024.

[202] J. Birch, "Animal sentience and the precautionary principle," *Animal Sentience*, vol. 2, no. 16, 2017.

[203] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *ICLR*, 2015.

[204] N. Srivastava et al., "Dropout: A simple way to prevent neural networks from overfitting," *JMLR*, vol. 15, pp. 1929–1958, 2014.

[205] K. He et al., "Deep residual learning for image recognition," in *CVPR*, 2016.

[206] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale (ViT)," in *ICLR*, 2021.

[207] I. Goodfellow et al., "Generative adversarial nets," in *NeurIPS*, 2014.

[208] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *NeurIPS*, 2020.

[209] R. Rombach et al., "High-resolution image synthesis with latent diffusion models," in *CVPR*, 2022.

[210] J. Schulman et al., "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[211] V. Mnih et al., "Human-level control through deep reinforcement learning," *Nature*, vol. 518, pp. 529–533, 2015.

[212] D. Silver et al., "Mastering the game of Go with deep neural networks and tree search," *Nature*, vol. 529, pp. 484–489, 2016.

[213] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, 2015.

[214] Y. Bengio, A. Courville, and P. Vincent, "Representation learning: A review and new perspectives," *IEEE TPAMI*, vol. 35, pp. 1798–1828, 2013.

[215] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[216] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, pp. 533–536, 1986.

[217] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, pp. 1735–1780, 1997.

[218] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *NeurIPS*, 2012.

[219] J. Deng et al., "ImageNet: A large-scale hierarchical image database," in *CVPR*, 2009.

[220] O. Russakovsky et al., "ImageNet large scale visual recognition challenge," *IJCV*, vol. 115, pp. 211–252, 2015.

[221] Dogster, "How many dogs are there? US & worldwide statistics," 2026. [Online]. Available: `https://www.dogster.com/statistics/how-many-dogs-are-there-statistics/`

[222] "Consumer acceptance of sustainable dog diets: A survey," 2025. PMC12560912 (reports a 2024 companion-dog population of ≈528 million).

[223] FEDIAF, "Facts & Figures 2025: European pet population," European Pet Food Industry Federation, 2025.

[224] AVMA, "2025 Pet Ownership and Demographics Sourcebook," American Veterinary Medical Association, 2025.

[225] Budget Direct, "Australian Dog and Cat Care Survey 2024," 2024.

[226] World Animal Foundation, "Pet ownership statistics 2026," 2026.

[227] B. F. Skinner, *Science and Human Behavior*. New York: Macmillan, 1953.

[228] K. Pryor, *Don't Shoot the Dog: The New Art of Teaching and Training*, rev. ed. New York: Bantam Books, 1999.

[229] J. P. Scott and J. L. Fuller, *Genetics and the Social Behavior of the Dog*. Chicago: University of Chicago Press, 1965.

[230] D. G. Freedman, J. A. King, and O. Elliot, "Critical period in the social development of dogs," *Science*, vol. 133, no. 3459, pp. 1016–1017, 1961.

[231] J. A. Serpell and D. L. Jagoe, "Early experience and the development of behaviour," in *The Domestic Dog: Its Evolution, Behaviour, and Interactions with People*, J. A. Serpell, Ed. Cambridge: Cambridge University Press, 1995, pp. 82–102.

[232] J. Topál, G. Gergely, A. Érdőhegyi, G. Csibra, and A. Miklósi, "Differential sensitivity to human communication in dogs, wolves, and human infants," *Science*, vol. 325, no. 5945, pp. 1269–1272, 2009.

[233] E. E. Bray, G. E. Gnanadesikan, D. J. Horschler, K. M. Levy, B. S. Kennedy, T. R. Famula, and E. L. MacLean, "Early-emerging and highly heritable sensitivity to human communication in dogs," *Current Biology*, vol. 31, no. 14, pp. 3132–3136, 2021.

[234] C. Mancini, "Animal–computer interaction: A manifesto," *ACM Interactions*, vol. 18, no. 4, pp. 69–73, 2011. DOI: 10.1145/1978822.1978836.

[235] W. B. Knox and P. Stone, "Interactively shaping agents via human reinforcement: The TAMER framework," in *Proc. K-CAP*, 2009, pp. 9–16.

[236] P. Christiano, J. Leike, T. B. Brown, M. Martic, S. Legg, and D. Amodei, "Deep reinforcement learning from human preferences," in *Proc. NeurIPS*, 2017.

[237] D. K. Lewis, *Convention: A Philosophical Study*. Cambridge, MA: Harvard University Press, 1969.

[238] L. Steels, "Evolving grounded communication for robots," *Trends in Cognitive Sciences*, vol. 7, no. 7, pp. 308–312, 2003.

[239] A. Lazaridou, A. S. Peysakhovich, and M. Baroni, "Multi-agent cooperation and the emergence of (natural) language," in *Proc. ICLR*, 2017.

[240] J. Foerster, I. A. Nye, and J. M. Whitehead, "Learning to communicate with deep multi-agent reinforcement learning," in *Proc. NeurIPS*, 2016.

[241] R. A. Gardner and B. T. Gardner, "Teaching sign language to a chimpanzee," *Science*, vol. 165, no. 3894, pp. 664–672, 1969.

[242] S. T. Savage-Rumbaugh and R. Lewin, *Kanzi: The Ape at the Brink of the Human Mind*. New York: Wiley, 1994.

[243] H. S. Terrace, L. A. Petitto, R. J. Sanders, and T. G. Bever, "Can an ape create a sentence?" *Science*, vol. 206, no. 4421, pp. 891–902, 1979.

[244] D. Britz, A. Goldie, M. Luong, and Q. Le, "Massive exploration of neural machine translation architectures," in *Proc. EMNLP*, 2017, pp. 1442–1451.

[245] M. Ott, S. Edunov, A. Baevski, A. Fan, S. Gross, N. Ng, D. Grangier, and M. Auli, "fairseq: A fast, extensible toolkit for sequence modeling," in *Proc. NAACL (Demos)*, 2019, pp. 48–53.

[246] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, et al., "Transformers: State-of-the-art natural language processing," in *Proc. EMNLP (Demos)*, 2020, pp. 38–45.

[247] A. Chowdhery, S. Narang, J. Devlin, M. Bosma, G. Mishra, A. Roberts, et al., "PaLM: Scaling language modeling with pathways," *Journal of Machine Learning Research*, vol. 24, no. 240, pp. 1–113, 2023.

[248] J. Wei, Y. Tay, R. Bommasani, C. Raffel, B. Zoph, S. Borgeaud, et al., "Emergent abilities of large language models," *Transactions on Machine Learning Research*, 2022.

[249] R. Schaeffer, C. Miranda, and S. Koyejo, "Are emergent abilities of large language models a mirage?" in *Proc. NeurIPS*, 2023.

[250] Y. Kong, J. Kim, and J. Bae, "HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis," in *Proc. NeurIPS*, 2020.

[251] Y. Ren, Y. Ruan, X. Tan, T. Qin, S. Zhao, Z. Zhao, and T.-Y. Liu, "FastSpeech: Fast, robust and controllable text to speech," in *Proc. NeurIPS*, 2019.

[252] A. Miklósi, *Dog Behaviour, Evolution, and Cognition*. Oxford: Oxford University Press, 2007.

[253] M. A. R. Udell, N. R. Dorey, and C. D. L. Wynne, "What did domestication do to dogs? A new account of dogs' sensitivity to human actions," *Biological Reviews*, vol. 85, no. 2, pp. 327–345, 2010.

[254] A. Horowitz, "Disambiguating the 'guilty look': Salient prompts to a familiar dog behaviour," *Behavioural Processes*, vol. 81, no. 3, pp. 447–452, 2009.

[255] S. Barocas and A. D. Selbst, "Big data's disparate impact," *California Law Review*, vol. 104, no. 3, pp. 671–732, 2016.

[256] S. Russell, *Human Compatible: Artificial Intelligence and the Problem of Control*. New York: Viking, 2019.

[257] D. Amodei, C. Olah, J. Steinhardt, J. Christiano, J. Schulman, and D. Mané, "Concrete problems in AI safety," arXiv:1606.06565, 2016.